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Does the Chilean water market price scarcity? Drought, spillovers and water rights transactions in Mediterranean Chile --Manuscript Draft--

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Corresponding Author:	Lisandro Roco Universidad San Sebastián
First Author:	Felipe Wolff
Order of Authors:	Felipe Wolff Lisandro Roco Jean Pierre Ometto Horacio Samaniego
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Opposed Reviewers:	

August 26th, 2026

Prof. Dr. Christopher Cvitanovic

Prof. Dr. Nicole Klenk

Editors-in-Chief

Environmental Science & Policy

Dear Editors,

We are pleased to submit *Does the Chilean water market price scarcity? Drought, spillovers and water rights transactions in Mediterranean Chile* for consideration as a research article in *Environmental Science & Policy*.

Chile's 1981 Water Code created the most thoroughly market-based system of water allocation in the world, on the premise that tradable rights would price scarcity and move water toward its highest-value use. That premise has been debated for four decades on largely institutional grounds. We test it directly, asking whether the price of a water right carries information about the physical availability of the water it entitles its holder to use.

It does, but weakly, at only one temporal scale, and in the wrong direction. Using 8,020 transactions across fifteen river basins between 2005 and 2014, we find that drought lowers rather than raises the price of an entitlement, because where a right is nominal its value tracks whether it can actually be exercised. The effect also reverses sign across basin boundaries, so that the regionally coherent drying described by climate projections produces a far weaker market response than a localised drought would. A market that devalues entitlements as water grows scarcer transmits no incentive to conserve it.

Set against this weak hydrological signal is a much stronger institutional one. An administrative declaration of water scarcity raises prices by about half within a month, roughly five times the aggregate effect of the drought index itself. The market responds more sharply to the announcement of scarcity than to the conditions the announcement describes, which locates price formation in an institutional event rather than

in physical conditions. We see this as speaking to the journal's interest in how the production and use of knowledge shapes decision making.

Chile is the canonical case for market-based water allocation and is watched wherever such systems are debated, so the findings bear on policy beyond the country studied. They also speak to a live reform, since our estimates characterise precisely the regime that Law 21,435 was written to correct and suggest it addresses the allocation of entitlements without repairing a price signal that moves the wrong way. The spatial specification is selected by likelihood ratio test rather than assumed, effects are decomposed following LeSage and Pace with uncertainty propagated by Monte Carlo, and the complete pipeline is publicly available as documented code at <https://github.com/horaciosamaniego/chile-water-rights-scarcity>.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have read and approved the submission and declare no competing interests.

Yours sincerely,

A handwritten signature in black ink, appearing to be 'Lisandro Roco', with a stylized, cursive script.

Dr. Lisandro Roco and Authors

lisandro.roco@uss.cl

Does the Chilean water market price scarcity? Drought, spillovers and water rights transactions in Mediterranean Chile

Highlights

- Chilean water markets were analyzed using a spatial model to understand price composition.
- Water prices respond weakly and unevenly to precipitation changes.
- Scarcity decrees have a much stronger effect on prices than drought conditions.
- Water prices may decline under SSP5-8.5, reducing incentives for conservation.

Does the Chilean water market price scarcity? Drought, spillovers and water rights transactions in Mediterranean Chile

Felipe Wolff

Center of Economics for Sustainable Development (CEDES), Faculty of Economics, Business and Government, Universidad San Sebastián, Valdivia 5090000, Chile.

Graduate School, Faculty of Sciences, Universidad Austral de Chile, Campus Isla Teja, Valdivia, Chile.

Lisandro Roco *

Center of Economics for Sustainable Development (CEDES), Faculty of Economics, Business and Government, Universidad San Sebastián, Valdivia 5090000, Chile.

Jean Pierre Ometto

Coordenação Geral de Ciências da Terra, Instituto Nacional de Pesquisas Espaciais (INPE), São José dos Campos, SP, Brazil.

Horacio Samaniego *

Laboratorio de Ecoinformática, Instituto de Conservación, Biodiversidad y Territorio, Universidad Austral de Chile, Campus Isla Teja, Valdivia, Chile.

Instituto de Sistemas Complejos de Valparaíso, Subida Artillería 470, Valparaíso, Chile.

*: Corresponding Authors E-mail: Lisandro Roco (lisandro.roco@uss.cl), Horacio Samaniego (horacio@ecoinformatica.cl)

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Abstract

Chile's 1981 Water Code established one of the world's most thoroughly market-based systems of water allocation, on the premise that tradable rights would price scarcity and move water toward its highest-value use. We examine whether prices in that market carry a hydrological signal, using 8,020 water rights transactions recorded in fifteen river basins of Mediterranean Chile between 2005 and 2014. A dynamic spatial Durbin model, selected against simpler spatial specifications by likelihood ratio, relates monthly transaction prices to market conditions and to a twelve-month Standardized Precipitation Index constructed from a forty-two year basin climatology. Prices respond to accumulated precipitation only at the annual scale, and the response reverses sign across basin boundaries. A one-unit improvement in the drought index raises prices by roughly a third where it occurs and lowers them by close to a fifth in neighboring basins, so that regionally synchronous drying produces a far weaker response than a localized one. Administrative signals dominate hydrological ones. A water scarcity decree in the preceding month raises prices by about half, some five times the aggregate effect of the drought index and acts only within the basin it names. Mining presence, in contrast, moves prices through neighboring basins rather than within the basin where the activity occurs. Prices transmit across basins with a lag rather than contemporaneously. Bias-corrected scenario projections imply that equilibrium prices fall by about a quarter by 2100 under SSP5-8.5 and remain unchanged under SSP1-2.6. The direction of that decline is robust across the estimated coefficient distribution, though its magnitude is not. A market that devalues entitlements as water grows scarcer transmits no incentive to conserve it, a failure the 2022 reform of the Water Code addresses only indirectly.

Keywords: Water markets, climate change, water management, spatial models.

JEL codes: Q25, Q54, P48, Q21

1. Introduction

Freshwater has been crucial for the development of social culture, leading societies to historically settle near water bodies (Rodda & Ubertaini, 2004). While premodern societies valued and protected water, economic and social developments have shifted their use towards a consumable good (UNEP & WMO, 1992). However, this perception is contested due to water's unique characteristics: irreplaceability, essential role in biotic cycles, scarcity, system-like properties, and its bulky nature, which complicates storage and trade (Savenije, 2002).

Climate change (CC) significantly disrupts atmospheric freshwater circulation, leading to altered precipitation patterns characterized by extreme events (Calvin et al., 2023;

Makarieva & Gorshkov, 2007; Sheil & Murdiyarso, 2009). These changes are expected to disproportionately reduce renewable water resources in dry subtropical regions, intensifying competition for water between human populations and ecosystems (Asadieh & Krakauer, 2015). Rural areas are likely to face severe freshwater scarcity, threatening food security and exacerbating social and economic tensions. Additionally, extreme rainfall events are projected to directly challenge water management and storage systems (McCornick, 2014).

Chile has been severely impacted by a megadrought. Between 1965 and 2019, four major drought events have been identified, underscoring the region's vulnerability to prolonged dry periods (Álvarez-Garretón et al., 2021). This large-scale phenomenon had profound consequences for ecosystems, the economy, and society. For instance, the reduction of glacial and snow cover has significantly limited streamflow during periods of high-water demand (World Bank, 2021). The OECD (2024) further emphasizes Chile's escalating water crisis in its 2024 Report, attributing it to a combination of high-water demand, erratic supply, and overallocation of resources. These factors illustrate a well-documented case of how CC exacerbates pressures on both human societies and ecosystems. The magnitude of the signal is evident in the record used here: mean annual precipitation across the studied basins fell from 1,053 mm in the 1980s to 739 mm over 2010–2020.

Compounding these challenges, water management in Chile is governed by the 1981 Water Law (Código de Aguas), which operates under a market-based model that inadequately addresses regional climatic variability. This regulatory shortfall has exacerbated social conflicts and intensified competition for water resources (Bauer, 2004a; Prieto, 2016). Furthermore, the rapid economic growth since the 1980s has placed additional strain on water resources. This has led to the overexploitation of aquifers (Fercovic et al., 2019; Aitken et al., 2016), raising concerns about the future stability of water markets in the face of CC.

The market-based design rests on a testable proposition. If tradable rights allocate water efficiently, then the price of a right should carry information about the physical availability of the resource it entitles its holder to use. Scarcity should be visible in prices, and prices should in turn discipline the use. This study asks whether that proposition holds in the Chilean case. Using water rights (WR) transactions from the Mediterranean zone, we estimate a dynamic spatial Durbin model that separates the influence of hydrological conditions from that of market dynamics, institutional signals and spillovers between basins, and we then ask what the estimated relationship implies under two contrasting climate futures. Three questions organize the analysis: whether WR prices respond to drought, at what temporal scale and spatial extent any such response operates, and whether the response is strong enough to matter relative to the market and administrative forces acting on the same prices. To the best of our knowledge this is the first assessment of the Chilean market that jointly identifies own-basin and neighboring-basin effects within a specification selected by test rather than assumed. Previous research has been implemented mostly at basin level and with limited incorporation of dynamics across time.

2. Background

2.1 Climate scenarios and water allocation

Human activities have caused global warming through the sustained release of greenhouse gases, producing shifts in the frequency and intensity of climatic extremes (Calvin et al., 2023). Global mean surface temperature is projected to exceed 1.5°C above pre-industrial levels under most scenarios by the end of the century, with the range across scenarios spanning 0.3°C to 4.8°C (Calvin et al., 2023). The spread across that range is not a measure of scientific uncertainty so much as of the socioeconomic trajectories the world may follow, which is why scenario analysis rather than single-path forecasting is the standard basis for assessing climate impacts.

Recent reports of The International Panel of Experts of Climate Change (IPCC) combine Representative Concentration Pathways, which describe trajectories of radiative forcing, with Shared Socioeconomic Pathways (Schipper et al., 2023). O'Neill et al. (2014) define SSPs as reference trajectories describing plausible societal development in the absence of climate policy. They are organized along two axes, the socioeconomic challenges to mitigation and those to adaptation, producing five narratives: SSP1 (sustainability) with low challenges on both axes, SSP2 an intermediate case, SSP3 (regional rivalry) high on both, SSP4 (inequality) low for mitigation and high for adaptation, and SSP5 (fossil-fueled development) high for mitigation and low for adaptation.

Applied to water, this framework is useful precisely because scarcity is not determined by supply alone. Water deficits depend on how scarce water is allocated among competing uses as much as on how much of it there is (Bijl et al., 2018), so scenarios that differ in socioeconomic development generate substantially different pictures of future water stress even under comparable hydrological change (Ndehedehe & Adeyeri, 2025). Coupling climate and socioeconomic pathways has accordingly become the standard basis for assessing future water availability and demand (Dau & Adeloye, 2021). That allocation dimension is here central as this study takes SSP1-2.6 and SSP5-8.5, which bracket the plausible range rather than representing a single best and worst case and does not strictly ask whether water will be available but what a market-based allocation mechanism does with the scarcity it faces.

2.2 Water scarcity and climate change

Accurately measuring the impact of CC is essential for designing effective mitigation strategies (Calvin et al., 2023). Since the late 1980s, progress has been made in public policies addressing freshwater scarcity (Booker et al., 2012). Loch (2013) demonstrated the profound interconnection between CC and freshwater management, emphasizing the need for policies and governance frameworks to align with this relationship. As future variations in water supply become increasingly likely, users at various governance levels will need to adapt, incurring associated costs. These challenges could be mitigated through the implementation of nature-based solutions (NBS) and by leveraging future opportunities for sustainable water management.

Efforts to address water scarcity have spurred innovative approaches that view water markets as potential opportunities (Grafton et al., 2012; Wheeler et al., 2017). However, Garrick et al. (2020) argue that conventional economic policy recommendations have fallen short, as they often overlook the complexities of water policies, and the distributive conflicts embedded in water allocation decisions. This is partly due to the failure to recognize

freshwater as an atypical market good. Furthermore, such approaches tend to neglect the ecological impacts of water management, focusing primarily on social and economic processes. In contrast, the adoption of NBS appears promising for enhancing freshwater availability, though their implementation remains limited and marginal to date (Vörösmarty et al., 2018).

In Chile, Barría et al. (2019) found that freshwater scarcity in the Perquilauquén River sub-basin is driven by a combination of CC effects and systemic issues such as over-allocation of WR, inadequate oversight, and the absence of mechanisms to adapt to mitigate climate variability. These factors are projected to further intensify water scarcity in the future. Similarly, Muñoz et al. (2020) concludes that the water crisis in Petorca River basin is primarily driven by CC and unsustainable water management practices. Compounding the problem, excessive WR allocation amounting to 18% of the basin's mean annual precipitation, have significantly worsened the crisis. Weak regulatory frameworks and overexploitation have resulted in severe water shortages, leaving communities at risk of losing access to drinking water.

2.3 The value of water

Economic theory posits that the efficient use of water as a resource requires pricing it at its long-term marginal cost, which reflects the additional benefit derived from its use in various economic activities. Such pricing would theoretically promote optimal resource allocation. During droughts, when demand exceeds supply, prices should rise to balance the two. However, this approach faces practical challenges, particularly due to environmental externalities that are often uncertain and difficult to quantify (Grafton et al., 2020; Schulz et al., 2024). Furthermore, issues such as WR allocation, governance inefficiencies, and social equity concerns complicate the implementation of a purely market-based solution. Despite these obstacles, global freshwater use has continued to rise at an annual rate of approximately 1% since the 1980s (UN, 2021), underscoring the urgent need for improving water management policies that reconcile economic efficiency with sustainability. Muñoz et al. (2020) highlight that a key aspect of water resource management is the reuse and proper management.

The prediction that drought raises the price of a water right holds only if the right reliably delivers water. Where entitlements are nominal and their exercise depends on physical availability, drought reduces the expected quantity of water attached to a right and may therefore reduce rather than raise its value. Which of these two forces dominates is an empirical question, and the answer characterizes whether a market transmits scarcity or merely records it.

WR have been advocated as an efficient mechanism for resource allocation, yet they face criticism for their multifaceted impacts (Grafton et al., 2016). Transaction costs, for instance, often impede effective and efficient exchanges under existing legal framework (Leonard et al., 2019; Loch et al., 2018). In Australia, Loch (2013) demonstrated that trading water allocations serves as a critical risk management tool during periods of scarcity, particularly during severe droughts. Grafton et al. (2016) further argue that water markets, when combined with effective planning and regulation, can yield significant economic,

environmental, and social benefits. Similarly, Arellano-Gonzalez et al. (2021) highlight that in California, more efficient allocation of water reductions could mitigate the costs of water scarcity, suggesting that institutional and market reforms present valuable opportunities for adaptation in dynamic and uncertain scenarios. Wang et al. (2025) compare the system implemented in China with the Chilean framework, showing that China's legal architecture remains incomplete and, despite strong government involvement, conflicts and tensions among water-market users persist.

2.4 The Chilean case

The Chilean market-based model to manage freshwater scarcity, regulate misuse by separating WR from land ownership (Prieto et al., 2019), and establishes a uniform management formula at national level, declaring water as a national good for public use while guaranteeing perpetual usage rights, granted free of charge, to those who register them. The General Directorate of Water (DGA) was granted authority to register WR, by assigning perpetual and freely transferable water usage rights on a first-come, first-served basis. Once assigned, owners may freely trade these WR (Donoso, 2018). The market was designed to reallocate water to its highest economic value by limiting demand through the transfer of rights and market forces, ensuring minimal public intervention and relying heavily on water user associations (Donoso, 2018). Evidence suggests that this management scheme has made WR more competitive where water is scarce (Herrera et al., 2019), but this uniform application across Chile happens without considering significant hydrological and productive differences between basins (Bauer, 2004a; Prieto, 2016).

Moreover, the original 1981 framework did not require rights holders to demonstrate actual, immediate use of the water they claimed, speculative entities accumulated vast portfolios of WR at zero initial cost, holding them merely to extract monopoly rents or to secure future industrial expansion. The resulting socio-economic landscape heavily favored agro-export conglomerates and transnational mining corporations, creating a deeply asymmetric playing field (Roose & Panez, 2020; Perez-Silva & Castillo, 2023). Additionally, the lack of hydrological information generated long-term overallocation of WR (Barría et al., 2019), with estimations around 30% of the national catchments (Reade Malagueño & D'Odorico, 2024).

The DGA is tasked with assisting in conflict resolution among water users, but without a dedicated budget (World Bank, 2011) which undermines its ability, leading to market distortions (Cristi et al., 2013). In 2005 the water law was reformed to address these challenges by introducing a non-use fee, though its effectiveness remains uncertain (Valenzuela et al., 2013). Subsequent regulations, including requirements for intended water use, the establishment of ecological flow rates (Bauer, 2015; Donoso, 2018), and public bidding in cases of high-water demand, have also proven largely ineffective (Peña-Torres et al., 2019). This is partly because WR issued prior to the reform remain unaltered, limiting the impact of these measures (Bauer, 2015). This has led to the overexploitation of aquifers and necessitated adjustments in administrative policies (Aitken et al., 2016; Fuster et al., 2024; Hernandez & Lobos, 2024).

Alongside the allocation regime, the DGA may declare a zone under water scarcity (*decreto de escasez hídrica*, SD hereinafter), a temporary administrative measure that acknowledges a deficit (DGA, 2025) and enables extraordinary management powers. These declarations are public, discrete and dated, which makes them an observable institutional signal of scarcity that is distinct from, though correlated with, the underlying hydrological conditions. The analysis below estimates the effect of each separately.

While there's evidence of progress in regulating Chile's water markets, significant challenges remain (Cristi et al., 2013). Transactions are conducted through private negotiations, where prices are determined by the information and bargaining power of the involved parties (Cristi et al., 2013; Donoso, 2018). Evidence indicates that water scarcity is driven by poor market and resource management, deregulation, a lack of territorial planning and integrated resource management, short-term decision-making, and widespread distrust among stakeholders (Babidge, 2016; Urquiza & Billi, 2020; Dardati & Sabat, 2024).

The framework was substantially amended by Law N° 21,435 (2022), which changed newly granted rights into renewable administrative concessions, prioritized human consumption, strengthened non-use forfeiture and DGA intervention, and restricted rights over glaciers and protected wetlands. Since the transactions analyzed predate the reform, our results characterize the former regime and should be interpreted as a diagnosis of the system that the reform sought to address.

3. Methods

3.1 Characterization of transactions

The study utilizes the mean transaction price of WR within river basins, derived from a comprehensive database of the Superintendence of Sanitary Services (SISS). This database encompasses over 15,000 WR transactions recorded nationwide between 1993 and 2014, and data are used to utilities tariffication processes. The subsample of recorded prices is used to estimate the value of raw water at basin level. Each transaction was geographically characterized by integrating it with relevant climatic and socioeconomic data. The transactions are documented at the municipal level and linked to specific water sources, allowing for accurate assignment to the corresponding river basins by the DGA.

The study focused on a selection of Chilean river basins characterized by a short range of climatic conditions and hydrological connectivity, enabling the analysis of climatic and spatial spillover effects across river basins. Basins were chosen based on the criterion of having recorded more than 30 WR transactions between 2005 and 2014. To consider direct hydrological connectivity, only basins situated between latitudes 30°S and 42°S were included in the analysis. Two basins meeting these criteria were subsequently excluded because the catchment precipitation record required to construct the drought index was unavailable for them. This selection process yielded 8,020 individual transactions across fifteen river basins, organized as a monthly panel of 1,785 basin-month observations spanning 2005–2014 (Figure 1).

The dataset revealed extended periods with no recorded transactions in most basins, and 60.4% of basin-months contain no transaction at all. Where a month records no transaction,

the price series carries the last observed value forward, so that a basin may hold an unchanged price for a long interval. The median basin holds a constant price for 49 consecutive months at its longest run. This is a property of the price series itself rather than an analytical choice, since a price can only be observed when a transaction occurs, but it has consequences for the estimated persistence of prices that must be confronted directly. We therefore report the persistence coefficient on three progressively more demanding subsamples in Section 4, and we treat the full panel estimate as an upper bound.

3.2 Construction of the drought index

Hydrological conditions were characterized using the Standardized Precipitation Index (SPI), which expresses accumulated precipitation as a probability transform onto the standard normal distribution (Alvarez-Garreton et al., 2018; Koudahe et al., 2017). Catchment-level monthly precipitation for the fifteen basins was obtained for 1979–2020, a forty-two-year record that satisfies the length requirement for a stable distributional fit and comfortably exceeds the World Meteorological Organization’s thirty-year minimum. The index was computed by accumulating precipitation over a window of k months, fitting a two-parameter gamma distribution to the strictly positive accumulations of each calendar month separately, and applying Thom’s mixed-distribution correction, $H(x) = q + (1 - q)G(x)$, where q is the empirical probability of zero precipitation. Fitting each calendar month separately is necessary in a strongly seasonal Mediterranean climate, and the zero correction is necessary because dry-season months frequently record no precipitation at all, in some basins in more than a fifth of peak summer months. Three accumulation windows were computed, at three, six and twelve months, and all three were carried through estimation. All are reported because the accumulation scale at which a market responds to drought is itself a result, and restricting attention to a single window would conceal it.

3.3 Development of the explanatory model

The dynamic Spatial Durbin Model (SDM) was employed to analyze the data. This model incorporates both temporal and spatial dimensions (Anselin, 1988; LeSage & Pace, 2009). The dataset was organized as a panel data within a matrix structure which allows us the explicit consideration of spatial dependencies and temporal dynamics, providing a robust basis for evaluating spillover effects across river basins (Elhorst, 2014). The SDM has recently been used to separate local from spillover effects in water resource systems, including regional supply suitability, watershed water security and water–energy–food efficiency (He et al., 2024; Zhang et al., 2024a, 2024b; Liu et al., 2025; Guo et al., 2026).

Spatial relationships between basins were represented by a row-standardized weight matrix $\mathbf{W}$ built from the four nearest neighboring basins, measured between basin centroids. The number of neighbors was selected primarily due to the limited number of basins retained after the initial filtering process. Because the panel is observed repeatedly over time, $\mathbf{W}$ is defined once at basin level and blocked across periods, so that each link connects distinct basins within the same month and no link connects a basin to itself in another month.

The estimated specification is:

$$y_t = \rho \mathbf{W} y_t + \tau y_{t-1} + \theta_\tau \mathbf{W} y_{t-1} + X_t \beta + \mathbf{W} X_t \theta + \mu + \iota_N \xi_t + \varepsilon_t \quad (1)$$

where y_t is the $N \times 1$ vector of log mean transaction prices across the fifteen basins in month t , W the row-standardized spatial weight matrix, ρ the spatial autoregressive parameter, τ the coefficient on the temporally lagged dependent variable, θ_τ the coefficient on its spatial lag, β the vector of coefficients on the basin-level covariates X_t , θ the vector of coefficients on the spatially lagged covariates WX_t , μ a vector of basin effects, ξ_t a period-specific effect, and ε_t the idiosyncratic error. Estimation is by maximum likelihood.

Equation (1) nests simple spatial forms and allows to set $\theta = 0$ and $\theta_\tau = 0$ to yield a spatial autoregressive model in which the ratio of indirect to direct effects collapses to $\rho/(1 - \rho)$ and is therefore identical for every covariate. The Durbin terms were retained or discarded based on a likelihood ratio test against that restriction rather than assumed. The test was repeated for each accumulation window of the drought index.

Effects were decomposed following LeSage and Pace (2009). For covariate k the multiplier matrix is $S_k(W) = (I - \rho W)^{-1}(\beta_k I + \theta_k W)$, whose mean diagonal gives the direct effect, whose mean row sum gives the total effect, and whose difference gives the indirect effect. Reporting the decomposition per covariate rather than as a single aggregate is essential, since the covariates are measured in incommensurable units and an average across them has no interpretation.

3.4 Variables

Table 1 reports the variables entering the model with their descriptive statistics over the estimation sample. The variables were constructed from the same transaction records and from public administrative sources. These comprise the number of transactions per basin-month, the mean flow transacted in liters per second, an indicator of mining presence in the basin, and an indicator for a SD in force in the previous month, together with the interaction between transacted flow and the SD. Prices are reflected in Chilean Pesos, but adjusted by UF (*Unidad de fomento*): UF is a inflation-indexed unit of account used in Chile to preserve the real value of financial obligations by adjusting daily to variations in the Consumer Price Index (Banco Central de Chile, 1989, art. 35, num. 9), to avoid the overdispersion of raw prices within and across basins. Mining presence is a binary indicator created by the relative weight of the mining activities in the basins, based on the percentage of income from mining patents, with a threshold of more than 2.5% of municipal GDP as suggested by Oyarzo and Paredes (2021).

SPI-12 averages -0.090 across the estimation window, so 2005–2014 was drier than the 1979–2020 climatology against which it is scored. This is consistent with the onset of the megadrought, and it sets the baseline from which the scenario projections in Section 4.3 depart.

3.5 Scenario evaluation

To assess the implications of projected CC for WR prices, the estimated model was evaluated under the SSP1-2.6 and SSP5-8.5 scenarios for four future periods: 2021–2040, 2041–2060, 2061–2080 and 2081–2100. Projected precipitation was obtained from the MIROC6 general circulation model via the IPCC WGI Atlas (Iturbide et al., 2022). Because SPI-12 is an annual accumulation and the projections are supplied as annual precipitation totals, the projected

and estimated indices are the same quantity and were produced by the same gamma transform against the same 1979–2020 basin climatology. Socioeconomic variables were held at their basin means over the estimation period, as there is insufficient basis for projecting changes in these variables.

Two corrections were applied. First, both scenarios sit approximately 32% below the observed climatology already in the first projected period, and they agree with one another there to within 1.7%. Since the scenarios have not yet diverged at that date, the shortfall identifies model dry bias rather than a climate signal and scoring raw model output against an observed-precipitation gamma would register that bias as permanent drought. Projections were therefore expressed as fractional change relative to the first projected period and applied to the observed basin climatology, following the standard delta-change approach. Second, prices were obtained as the long-run equilibrium of equation (1) rather than as a single-period linear prediction. Imposing $y_t = y_{t-1}$ gives:

$$y^* = [(1 - \tau)\mathbf{I} - (\rho + \theta_\tau)\mathbf{W}]^{-1} (\alpha + Z) \quad (2)$$

where Z collects every covariate other than the lagged price together with its spatial lag. Uncertainty was propagated by drawing 4,000 parameter vectors from the estimated coefficient distribution and solving (2) for each, discarding any draw for which no stable steady state exists. Results are reported as medians with 90% intervals.

4. Results and Discussion

Table 2 reports the estimated model in both its autoregressive and the Durbin form. The likelihood ratio test rejects the autoregressive restriction decisively (LR = 33.35 on 7 degrees of freedom, $p < 0.0001$), and the Akaike criterion agrees, so the Durbin specification is retained, and interpretation proceeds from it. Under the autoregressive restriction the ratio of indirect to direct effects is forced to take the same value for every covariate, which would have concealed the sign reversal reported below.

Prices respond to accumulated precipitation, but only at the annual scale (Figure 2). The three- and six-month accumulation windows produce coefficients indistinguishable from zero in both specifications, while the twelve-month window is significant in both (0.102, $z = 2.34$ in the autoregressive form and 0.290, $z = 2.85$ in the Durbin form). The opposition between own-basin and neighboring-basin responses discussed below likewise appears only at the annual window. A WR is a durable asset whose value should track multi-season availability rather than recent weather, and the estimated pattern is consistent with that reading. It also indicates that studies relying on short-window drought indices may report null results that reflect the choice of window rather than the absence of a relationship.

The own-basin effect is positive. A one-unit improvement in SPI-12 raises prices in the basin where it occurs by roughly a third, so drought lowers them, which inverts the intuition that scarcity raises price. Section 2.3 anticipated this outcome: where an entitlement is nominal and its exercise depends on physical availability, drought reduces the expected quantity of water attached to the right, and the asset devalues accordingly. The market records scarcity rather than transmitting a scarcity premium.

Administrative signals act far more strongly than hydrological ones. A SD in force in the previous month raised prices by about half (0.426, $z = 4.74$), an effect five times (*i.e.*, 0.475 vs. 0.087) the magnitude of a one-unit change in the drought index and far more precisely estimated. The interaction between transacted flow and the decree is negative and highly significant (-0.025 , $z = -5.42$), indicating that larger transactions attract proportionally lower unit prices under declared scarcity. Market variables behave as expected, with the number of transactions raising prices (0.009, $z = 4.14$) and larger mean flows per transaction lowering them (-0.0009 , $z = -3.99$).

Contemporaneous price spillovers are absent. The spatial autoregressive parameter is not significantly different from zero in the Durbin specification ($\rho = -0.032$, $z = -0.86$), while the spatial lag of the previous month's price is significant and positive (0.112, $z = 2.68$). Prices therefore transmit between basins with a delay rather than simultaneously, which describes a diffusion process across space and time rather than a spatial equilibrium. The apparent contemporaneous dependence in the autoregressive specification ($\rho = 0.079$, $z = 3.52$) does not survive the inclusion of the Durbin terms, which is a further reason to prefer the general specification.

4.1 Direct, indirect and total effects

Figure 3 reports the effects of decomposition. Four covariates have spillover effects distinguishable from zero, and in two of these the spillover opposes the own-basin effect in sign. Neither pattern can arise under the autoregressive restriction, which forces the ratio of indirect to direct effects to equal $\rho/(1 - \rho)$ for every covariate alike, here 0.086. The observed ratios instead range from -1.05 to 8.46 .

The drought index carries a positive direct effect of 0.291 and a negative indirect effect of -0.203 , so wetter conditions in a basin raise its prices while wetter conditions among its neighbors lower them. Both intervals exclude zero, which deserves comment because the spatially lagged coefficient from which the indirect effect derives does not reach conventional significance on its own (Table 2). Inference on effects and inference on coefficients need not agree, since the effect combines ρ and θ and the two covary, and it is the effect rather than the coefficient that carries the economic interpretation. The asymmetry is readily interpreted as substitution: when water is physically available nearby, buyers hold alternatives, and local entitlements lose scarcity value. It also implies that regionally synchronous drying, which is what climate projections describe, produces a far weaker aggregate response than a localized drought would, since the two channels largely offset one another. The total effect of 0.087 is an order of magnitude smaller than the direct effect.

Mining presence shows the opposite configuration. Its direct effect of 0.055 cannot be distinguished from zero, while its indirect effect of 0.480 is estimated with an interval well clear of it, so extractive activity in adjacent basins moves prices where extractive activity within the basin does not. Demand pressure from mining is evidently transmitted through the market rather than confined to the basin where the activity is located, which bears on the recurring proposal to manage water at basin scale in isolation.

The number of transactions has direct and indirect effects that almost exactly cancel, at 0.010 and -0.010, and both intervals exclude zero. The offsetting is therefore a property of the estimates rather than a consequence of imprecision. Market activity redistributes value between basins without generating any at the regional scale, a result invisible in any specification that constrains the two effects to share a sign.

The SD behaves differently again. Its direct effect is large and precisely estimated, while its spillover cannot be distinguished from zero, so an administrative declaration moves prices in the basin it names and leaves neighboring basins largely unaffected. The market reads the decree as information about a particular basin rather than as a signal about regional conditions.

4.2 Robustness of the persistence estimate

The lagged price carries the largest coefficient in the model, and because 60.4% of basin-months record no transaction the price series is partly a carried-forward one. Persistence estimated on such a series is mechanically inflated. Regressing price on its own lag gives 0.823 on the full panel, 0.715 when restricted to months in which a transaction was observed, and 0.582 when restricted to consecutive observed months. Persistence is therefore real but overstated by the structure of the record, and the full-panel figure should be read as an upper bound. The substantive results are unaffected, since the drought, decree and spillover coefficients are estimated from variation across basins and months rather than from the persistence channel.

Including maximum monthly precipitation alongside SPI-12 leaves the drought result intact and slightly strengthens it (0.330, $z = 3.15$), while maximum precipitation itself is not significant in the Durbin specification. The two are alternative measures of the same underlying phenomenon, and the accumulated index is the theoretically appropriate one, so the reported specification retains SPI-12 alone.

4.3 Model evaluation in the climate change context

Under SSP1-2.6, projected precipitation is essentially unchanged across the century and equilibrium prices are correspondingly flat, varying by less than two percent in any period. Under SSP5-8.5, precipitation declines by roughly a quarter by 2081–2100 and prices fall by 22.8% (Figure 4). The direction is robust, with 97% of draws yielding a decline, but the magnitude is uncertain by a factor of nine, the 90% interval for the final period running from -39.6% to -4.3%.

The projected decline is smaller than the underlying precipitation change alone would imply, and it runs downward. The first follows from the substitution effect identified in Section 4.1. Climate projections describe regionally coherent drying, and under such drying the positive own-basin effect and the negative neighboring-basin effect largely cancel, so a market that would respond sharply to a localized drought responds weakly to a general one. The second is more consequential. Under the most severe scenario considered, WR become less valuable as water becomes scarcer, which is the opposite of what a functioning scarcity-pricing mechanism would produce.

While the model has limitations in projecting prices, it serves as an indicator of projected scarcity under the most severe CC scenarios. According to Valdés-Pineda et al. (2014), a critical issue in Chile is that basins with the highest water demand are those with the largest populations and significant mining activity, primarily located in central and northern Chile. This has sparked discussions about exporting water from southern basins. While such measures could alleviate resource pressure, they fail to account for other critical factors, such as ecosystem services related to regulation and provision, as well as impacts on ecosystems, biodiversity, species invasions, and system stability. Our finding that mining presence acts primarily through neighboring basins rather than within them adds a further consideration to that debate.

In addition to unaddressed ecological effects, governance challenges arise, as changes in projected precipitation may affect WR prices through both local market conditions and spatial interactions among neighboring basins. Traditional solutions are clearly insufficient to address these complex issues; instead, innovation and a focus on the root problem—water use rather than mere provision or scarcity—are required (Lopez-Gunn & Llamas, 2008).

4.4 Policy implications and limitations

The general lack of information on WR transactions introduces biases for model development. Previous national-level analyses have shown that socioeconomic characteristics of a basin can influence WR prices (Araos & Roco, 2025). This study estimates the effects of drought and spatial spillovers between basins, revealing that neither operates in the direction the market-based framework assumes. This underscores the need for policies and territorial planning to adopt a holistic, system-wide perspective, as changes in one basin may affect others through market and spatial dynamics.

Most studies focus on determinants of WR price transactions without considering spatial variation or inter-basin effects. This study provides evidence of the influence of neighboring basins and climatic factors, demonstrating that homogeneous formulas are inadequate.

Currently, the minimum ecological flow required to maintain WR, as mandated by the government, corresponds to 20% of the river basin's mean flow (MMA, 2012). This policy applies to WR established after that date, while maintaining minimum flows for rights granted earlier. However, this policy overlooks significant ecological differences among basins, which have evolved over time (e.g., during drought periods), and fails to account for varying minimum flow requirements needed to sustain ecosystems. Legal and institutional frameworks play a critical role in shaping transaction costs and market functionality. Evidence from Colorado indicates that legal uncertainty and third-party injury rules can substantially raise transaction costs, even in active water markets (Womble & Hanemann, 2020). Chile's water market may be transitioning into a phase like the 'fifth stage' of water management, as proposed by Loch et al., characterized by heightened uncertainty and the need for adaptive governance mechanisms in response to socio-environmental pressures (Loch et al., 2020).

SD constitute an important state intervention for managing water during periods of severe hydrological stress. In Chile, their relevance is heightened by persistent drought and spatially uneven water availability, as well as by structural over-allocation and localized

overexploitation of water resources (Garreaud et al., 2017, 2020, 2025; Barría et al., 2019; Taucare et al., 2024). In this context, SD can influence the effective allocation and redistribution of water among competing users, with potentially significant socioeconomic and spatial implications within Chile's market-based water governance system (Bauer, 1998, 2004a, 2004b; Donoso, 2015). The strength of the SD effect relative to the drought effect carries a policy reading of its own. The market responds far more sharply to an administrative declaration of scarcity than to the hydrological conditions that declaration describes. Where this reflects the informational value of a public signal in a market conducted through private negotiation, it suggests that improving the timeliness and granularity of hydrological information could substantially improve price formation. Where instead it reflects the regulatory consequences that follow a declaration, it suggests that the market is pricing administrative risk rather than water. Distinguishing the two is beyond what the present data allow, but the distinction is consequential for the design of the concession regime introduced by Law 21,435.

Environmental flows represent the specific flow regime necessary to maintain a river's aquatic habitats and ecosystem processes (UN, 2021). Identifying the ecosystem services provided by a basin and valuing them allows the determination of a minimum hydrological regime required to deliver those services. This minimum becomes the flow maintenance value for the basin, and recovery actions are necessary for basins that have fallen below this threshold. NBS can be applied in such cases to reduce operational and capital costs related to water or to enhance productivity in drought-affected basins (Abell et al., 2017; Garrick et al., 2017; Horne et al., 2017). NBS, combined with water capture management, can increase groundwater recharge, conserve surface water, enhance soil moisture retention, and improve regional precipitation management (UN, 2021).

The past and forthcoming reforms of Chile's Water Code are highly relevant for understanding the evolution of water governance and the functioning of water markets. The 2005 reform sought to address speculative WR accumulation through the introduction of non-use fee, while the 2022 reform strengthened state intervention by introducing stricter conditions on the use and potential expiration of WR (Donoso, 2015; Perez-Silva & Castillo, 2023; Adiego et al., 2025). These reforms, together with forthcoming regulatory changes, are particularly important in a context of persistent water scarcity and structural over-allocation, as they may significantly affect the incentives to hold, trade, and use WR and, consequently, the spatial and socioeconomic distribution of water resources.

Public policies must be adjusted to reflect the specific characteristics of each river basin, with NBS serving as a viable alternative to maintain ecological flows in each unit (Arellano-Gonzalez et al., 2021). Critiques of the WR transaction system highlight its focus on water use rather than the preservation of water sources. However, similar studies emphasize that regulating water use types and improving efficiency should be prioritized (Valdés-Pineda et al., 2014). In markets with structural similarities to the Chilean case, such as Australia, the composition of agricultural activity—particularly the prevalence of permanent crops—has been shown to influence the direction of water allocation flows under scarcity conditions (Zhao et al., 2024). In the absence of formal regulatory frameworks, cooperative mechanisms among agricultural users—such as those observed in northern Chile—can emerge as effective alternatives. These cooperatives implement static redistribution rules to manage

allocation uncertainty, a phenomenon also modeled in the semi-arid north of Chile (Delorit & Block, 2018).

This approach does not contradict the principles of water use and markets but rather enables sustainable management of this scarce and irreplaceable resource. Traditionally, the economic value of water sources has been determined by use and extraction costs. However, this narrow perspective limits analysis and overlooks potential efficiency measures that could be implemented post-extraction. According to Abell et al. (2017), Garrick et al. (2017), and Horne et al. (2017), investment in river basin conservation could yield significant returns.

McNamara et al. (2021) emphasize that the hydrological responses to CC, land-use pressures, and irrigation expansion vary substantially across basins, underscoring that effective adaptation requires basin-specific planning rather than uniform national prescriptions. This can be addressed by the development of NBS as an alternative for recharge and the establishment of mechanisms to improve water use efficiency and control. These measures are essential for ensuring the long-term sustainability of water resources in Chile.

The management of water in Chile reflects a critical decoupling between biophysical dynamics and its institutional framework. By treating WR as commodities decoupled from land and ecosystem boundaries, the legal architecture ignores the continuous, interconnected nature of river basins. This rigid administrative approach fails to adapt to climate-driven hydrological changes—such as severe droughts and shifting snowpacks—creating a structural disconnect where static legal entitlements conflict directly with the dynamic ecological realities of water scarcity.

Several limitations qualify these results. The transaction data do not distinguish among right types with systematically different prices, while the SISS price denominator cannot be fully verified. The 60% of basin-months without transactions limits inference on short-run dynamics, and the small number of basins reduces the reliability of spatial inference. The scenario exercise relies on a single climate model, fixed socioeconomic conditions, and a conservative delta correction. Most importantly, it reflects the 2005–2014 regulatory regime and therefore should not be interpreted as a forecast under the current regime following Law 21,435. Additional variables such as land use, industrial activity, and water use could improve robustness, but their limited temporal and spatial resolution would require interpolation and potentially introduce bias.

5. Conclusions

This study asked whether the Chilean water market prices scarcity, as the 1981 Water Code assumed it would. The answer is that it does so weakly, at only one temporal scale, and in the wrong direction. Prices respond to accumulated precipitation over twelve months but not over three or six, and the response indicates that drought lowers rather than raises the value of a water right. A market designed to allocate water efficiently under scarcity devalues the entitlement precisely when the resource becomes scarcer and therefore transmits no incentive to conserve it.

The spatial structure reinforces this reading. Wetter conditions in neighboring basins depress local prices by nearly as much as wetter conditions locally raise them, so that the regionally coherent drying described by climate projections produces a much weaker aggregate response than a localized drought would. Contemporaneous price spillovers between basins are absent, though prices do transmit with a lag, and demand pressure from mining acts principally through neighboring basins rather than within the basin where the activity occurs. Homogeneous, basin-by-basin regulation is poorly matched to a market that behaves this way.

Against these hydrological effects stands a much stronger institutional one. An administrative declaration of water scarcity raises prices by about half within a month, several times the magnitude of any climate variable in the model. The market responds to the announcement of scarcity far more than to scarcity itself. Whether this reflects the informational value of a public signal in a market conducted through private negotiation, or the pricing of regulatory risk, it indicates that price formation in this market is governed by institutional events rather than by physical conditions.

Scenario projections carry these estimates forward. Under aggressive mitigation, equilibrium prices are essentially unchanged across the century. Under high emissions, they fall by roughly a quarter by 2100, with the direction robust and the magnitude uncertain. These projections describe the regime that Law 21,435 was written to reform, and they support the direction of that reform while suggesting it does not go far enough. Converting perpetual rights into renewable concessions addresses the allocation of entitlements, but it does not by itself repair a price signal that moves the wrong way. Achieving that would require attaching value to the maintenance of the resource rather than to the entitlement alone, through environmental flow requirements that reflect basin-specific ecological conditions, through greater transparency in transaction records, and through recognition that basins are connected by a market even where they are not connected by a river.

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Data and code availability

The analysis pipeline, comprising the construction of the drought index, the spatial weight matrix, model estimation, the effects decomposition and the scenario projection, is available as documented Python scripts here: <https://github.com/horaciosamaniego/chile-water-rights-scarcity/>. Data are available on request to authors.

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Figure 1. Water right transactions. **A)** Total water right transactions per year between 2005 and 2014. **B)** Total water right transactions per river basin. Colors show the location of selected river basins in the map.

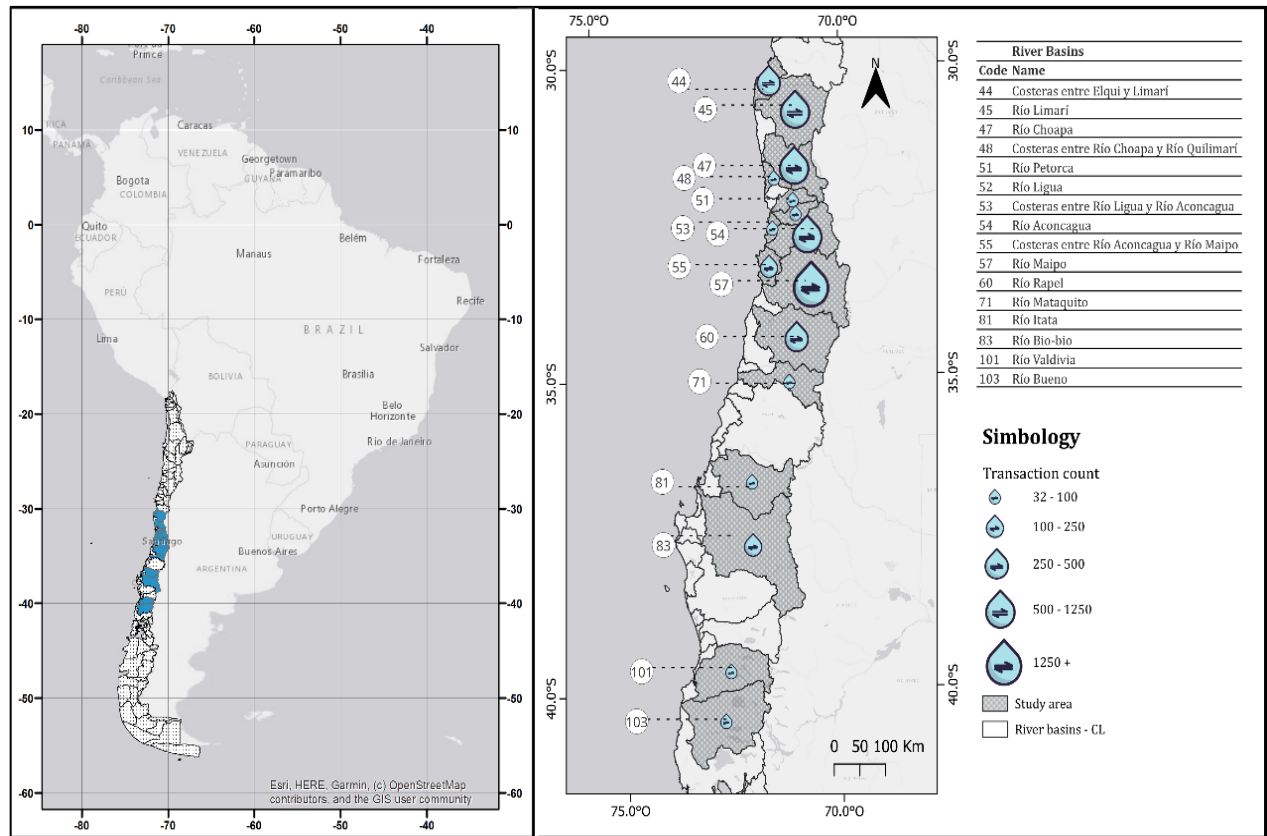


Figure 2. Response of water rights prices to drought by precipitation accumulation window. Own-basin and neighboring-basin coefficients on the Standardized Precipitation Index at three, six and twelve-month accumulation, estimated from the spatial Durbin specification, with 95% confidence intervals. Prices respond only at the annual scale, and the own-basin and neighboring-basin responses take opposite signs at that scale alone.

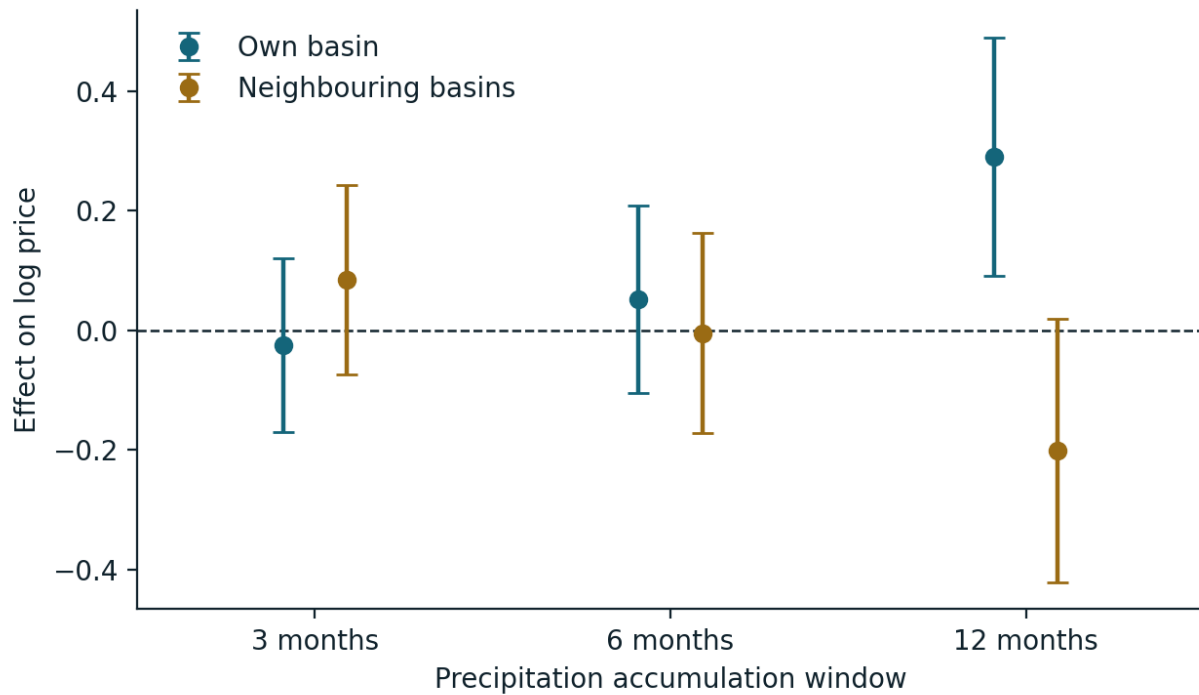


Figure 3. Direct and indirect effects of each covariate on water rights transaction prices. **(A)** Covariates whose effects are resolvable on the full scale. **(B)** The remaining covariates on a horizontal axis magnified roughly eightfold, so marker positions are not comparable between panels. Effects were computed from the spatial Durbin estimates in Table 2 following LeSage and Pace (2009), with medians and 90% intervals from 4,000 draws of the estimated coefficient distribution. Direct effects operate within the basin and indirect effects through neighboring basins. Filled markers denote intervals excluding zero, open markers intervals spanning it. Under the autoregressive restriction the two effects would be constrained to share a sign in every row, at a fixed ratio of $\rho/(1 - \rho) = 0.086$.

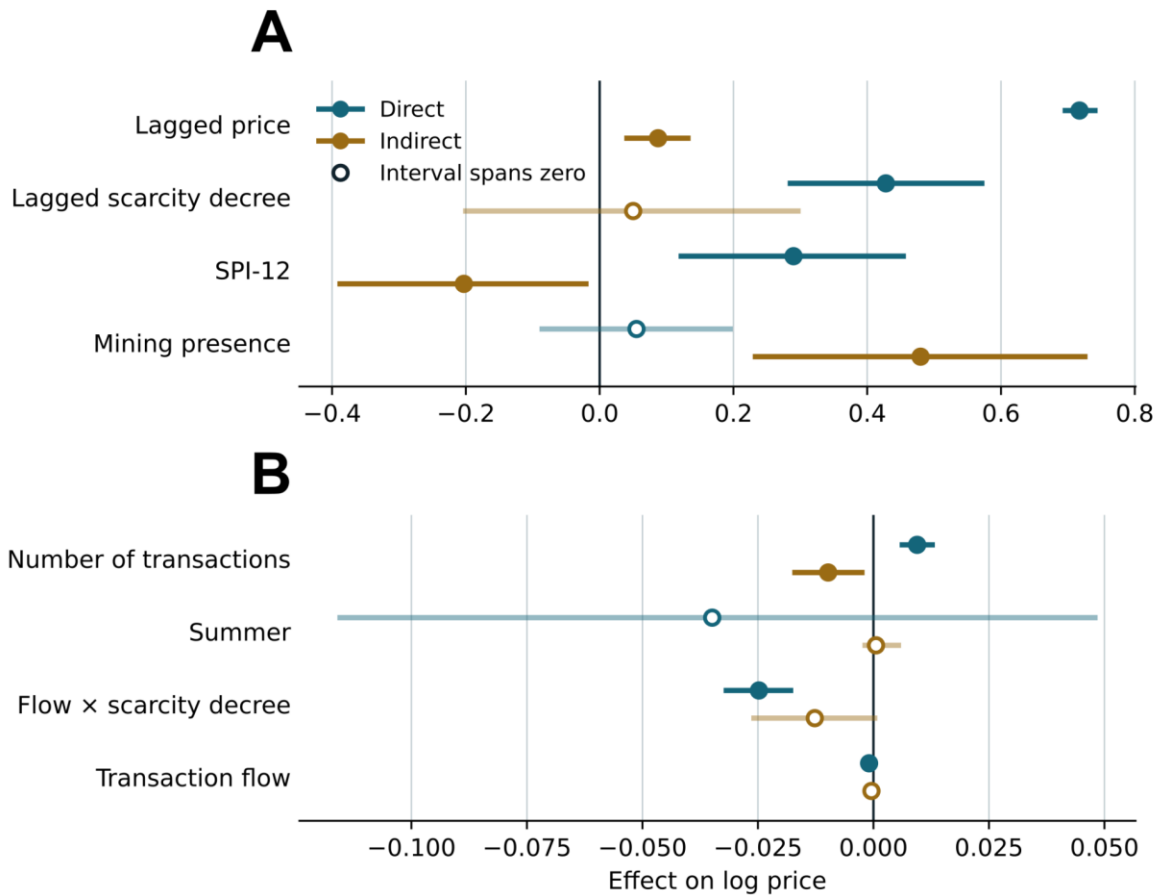


Figure 4. Projected change in long-run equilibrium water rights prices relative to the first projected period, under SSP1-2.6 and SSP5-8.5. Shaded bands show 90% intervals from 4,000 draws of the estimated coefficient distribution.

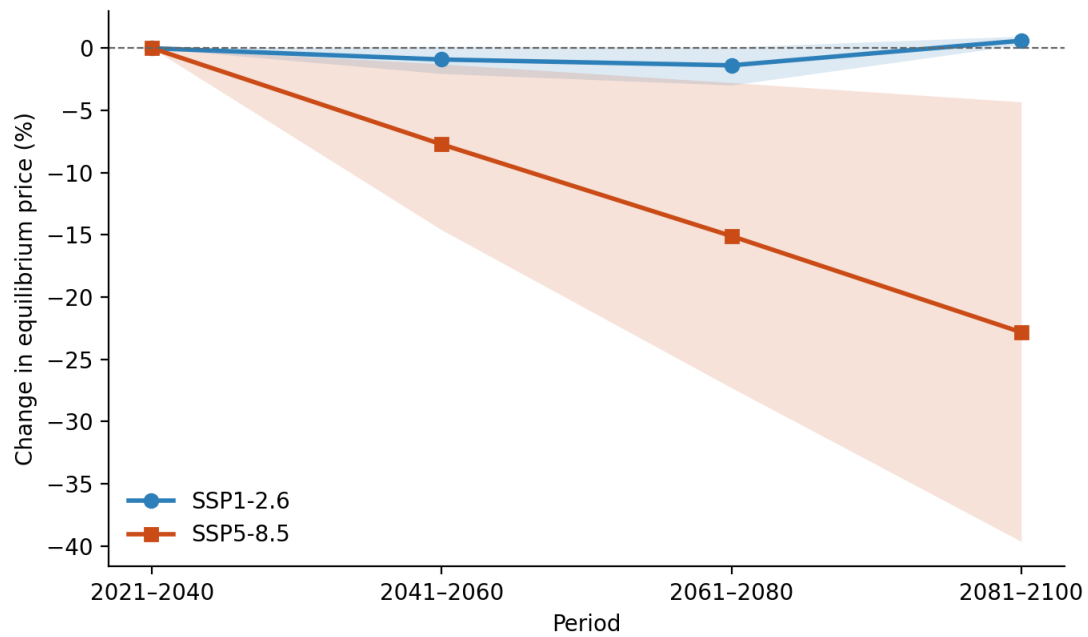


Table 1. Variables entering the model, with descriptive statistics over the estimation sample. Fifteen river basins of Mediterranean Chile observed monthly over 2005–2014, giving $N = 1,785$ basin-months after the loss of the first period to the lagged price. Prices are logarithms of the mean transaction price in UF. SPI-12 is the twelve-month Standardized Precipitation Index computed from catchment precipitation over 1979–2020, so its negative sample mean indicates that the estimation decade was drier than the reference climatology. Indicator variables take the value 0 or 1, and their means give the share of basin-months in the relevant state.

Variable	Description	Mean	Standard Deviation	Range
<i>Price</i>	Logarithm of mean transaction price (UF)	4.379	1.907	–2.98 to 12.82
<i>Lagged price</i>	Price in the previous month	4.370	1.904	–2.98 to 12.82
<i>Transaction flow</i>	Mean flow transacted per basin-month (l/s)	21.714	107.140	0.01 to 1400.0
<i>Number of transactions</i>	Transactions per basin-month	4.493	12.401	0 to 88
<i>Mining presence</i>	Indicator of mining activity in the basin	0.158	0.365	0 or 1
<i>Lagged SD</i>	Water scarcity decree in force, previous month	0.159	0.365	0 or 1
<i>Flow × SD</i>	Interaction of transacted flow and decree	0.917	5.671	0 to 129.2
<i>Summer</i>	Indicator for the austral summer half-year	0.496	0.500	0 or 1
<i>SPI-12</i>	Twelve-month Standardized Precipitation Index	–0.090	0.604	–2.03 to 1.71

Table 2. Determinants of water rights transaction prices in fifteen river basins of Mediterranean Chile, 2005–2014. Maximum likelihood estimates of eq. (1). The dependent variable is the logarithm of the mean transaction price in UF. Own-basin terms appear in the upper panel and spatially lagged terms in the lower. The SAR column imposes $\theta = 0$ and $\theta \sim \tau \sim 0$, so no spatially lagged covariates enter and dashes mark terms that specification excludes; the SDM column is unrestricted. $\mathbf{W}$ is row-standardized over the four nearest neighboring basin centroids and blocked across periods, so every link connects distinct basins within the same month. $\mathbf{W} \times \text{price}$ is the spatial autoregressive parameter ρ . A likelihood ratio test rejects the SAR restriction (LR = 33.35, df = 7, $p < 0.001$), and interpretation proceeds from the SDM column.

Variable	SAR				SDM			
	est.	s.e.	z	p	est.	s.e.	z	p
<i>Constant</i>	0.719	0.106	6.81	<0.001	0.792	0.138	5.73	<0.001
<i>Transaction flow</i>	-0.001	0.0001	-3.81	<0.001	-0.001	0.0001	-3.99	<0.001
<i>Summer</i>	-0.029	0.050	-0.59	0.557	-0.035	0.050	-0.70	0.482
<i>Mining presence</i>	0.126	0.085	1.49	0.136	0.060	0.086	0.69	0.488
<i>Number of transactions</i>	0.008	0.002	3.51	<0.001	0.009	0.002	4.14	<0.001
<i>Flow $\times$ SD</i>	-0.025	0.005	-5.50	<0.001	-0.025	0.005	-5.42	<0.001
<i>Lagged price</i>	0.743	0.015	48.42	<0.001	0.718	0.016	44.77	<0.001
<i>Lagged SD</i>	0.486	0.088	5.52	<0.001	0.426	0.090	4.74	<0.001
<i>SPI-12</i>	0.102	0.044	2.34	0.020	0.290	0.102	2.85	0.004
<i>W $\times$ transaction flow</i>	—	—	—	—	-0.001	0.001	-0.71	0.478
<i>W $\times$ mining presence</i>	—	—	—	—	0.496	0.154	3.23	0.001
<i>W $\times$ number of transactions</i>	—	—	—	—	-0.010	0.004	-2.02	0.043
<i>W $\times$ flow $\times$ SD</i>	—	—	—	—	-0.014	0.008	-1.64	0.100
<i>W $\times$ lagged price</i>	—	—	—	—	0.112	0.042	2.68	0.007
<i>W $\times$ lagged SD</i>	—	—	—	—	0.064	0.156	0.41	0.681
<i>W $\times$ SPI-12</i>	—	—	—	—	-0.200	0.112	-1.78	0.075
<i>W $\times$ price (ρ)</i>	0.079	0.022	3.52	<0.001	-0.032	0.037	-0.86	0.389
<i>Log likelihood</i>	-2,620				-2,603			
<i>Akaike criterion</i>	5,261				5,241			
<i>Pseudo R²</i>	0.697				0.702			
<i>Observations</i>	1,785				1,785			

Supplementary Material

TABLE A1. Decomposition of the estimated effects into own-basin, spillover and total components. Direct, indirect and total effects computed from the spatial Durbin estimates in Table 2 following LeSage and Pace (2009). For covariate k the multiplier matrix is $S \sim k \sim (\mathbf{W}) = (\mathbf{I} - \rho \mathbf{W})^{-1} (\beta \sim k \sim \mathbf{I} + \theta \sim k \sim \mathbf{W})$; the direct effect is its mean diagonal, the total effect its mean row sum, and the indirect effect the difference. Effects are expressed in log price units. The final column reports the ratio of indirect to direct effects, which under the autoregressive restriction would equal $\rho/(1 - \rho)$ for every covariate and take the value 0.086. That the ratio instead varies from -1.05 to 8.46 and changes sign is a property of the Durbin specification.

Variable	Direct	90% Interval	Indirect	90% Interval
Lagged price	0.717	0.692 to 0.744	0.087	0.037 to 0.136
Lagged SD	0.428	0.281 to 0.575	0.050	-0.204 to 0.301
SPI-12	0.290	0.118 to 0.458	-0.203	-0.392 to -0.017
Mining presence	0.055	-0.090 to 0.199	0.480	0.229 to 0.729
Number of transactions	0.0095	0.006 to 0.013	-0.0098	-0.018 to -0.002
Summer	-0.035	-0.116 to 0.049	0.001	-0.002 to 0.006
Flow $\times$ SD	-0.025	-0.032 to -0.017	-0.013	-0.026 to 0.001
Transaction flow	-0.0009	-0.0013 to -0.0005	-0.0004	-0.0015 to 0.0006

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Does the Chilean water market price scarcity? Drought, spillovers and water rights transactions in Mediterranean Chile

Felipe Wolff

Center of Economics for Sustainable Development (CEDES), Faculty of Economics, Business and Government, Universidad San Sebastián, Valdivia 5090000, Chile.

Graduate School, Faculty of Sciences, Universidad Austral de Chile, Campus Isla Teja, Valdivia, Chile.

Lisandro Roco *

Center of Economics for Sustainable Development (CEDES), Faculty of Economics, Business and Government, Universidad San Sebastián, Valdivia 5090000, Chile.

Jean Pierre Ometto

Coordenação Geral de Ciências da Terra, Instituto Nacional de Pesquisas Espaciais (INPE), São José dos Campos, SP, Brazil.

Horacio Samaniego *

Laboratorio de Ecoinformática, Instituto de Conservación, Biodiversidad y Territorio, Universidad Austral de Chile, Campus Isla Teja, Valdivia, Chile.

Instituto de Sistemas Complejos de Valparaíso, Subida Artillería 470, Valparaíso, Chile.

*: Corresponding Authors E-mail: Lisandro Roco (lisandro.roco@uss.cl), Horacio Samaniego (horacio@ecoinformatica.cl)

Does the Chilean water market price scarcity? Drought, spillovers and water rights transactions in Mediterranean Chile

Abstract

Chile's 1981 Water Code established one of the world's most thoroughly market-based systems of water allocation, on the premise that tradable rights would price scarcity and move water toward its highest-value use. We examine whether prices in that market carry a hydrological signal, using 8,020 water rights transactions recorded in fifteen river basins of Mediterranean Chile between 2005 and 2014. A dynamic spatial Durbin model, selected against simpler spatial specifications by likelihood ratio, relates monthly transaction prices to market conditions and to a twelve-month Standardized Precipitation Index constructed from a forty-two year basin climatology. Prices respond to accumulated precipitation only at the annual scale, and the response reverses sign across basin boundaries. A one-unit improvement in the drought index raises prices by roughly a third where it occurs and lowers them by close to a fifth in neighboring basins, so that regionally synchronous drying produces a far weaker response than a localized one. Administrative signals dominate hydrological ones. A water scarcity decree in the preceding month raises prices by about half, some five times the aggregate effect of the drought index and acts only within the basin it names. Mining presence, in contrast, moves prices through neighboring basins rather than within the basin where the activity occurs. Prices transmit across basins with a lag rather than contemporaneously. Bias-corrected scenario projections imply that equilibrium prices fall by about a quarter by 2100 under SSP5-8.5 and remain unchanged under SSP1-2.6. The direction of that decline is robust across the estimated coefficient distribution, though its magnitude is not. A market that devalues entitlements as water grows scarcer transmits no incentive to conserve it, a failure the 2022 reform of the Water Code addresses only indirectly.

Keywords: Water markets, climate change, water management, spatial models.

JEL codes: Q25, Q54, P48, Q21

1. Introduction

Freshwater has been crucial for the development of social culture, leading societies to historically settle near water bodies (Rodda & Ubertaini, 2004). While premodern societies valued and protected water, economic and social developments have shifted their use towards a consumable good (UNEP & WMO, 1992). However, this perception is contested due to water's unique characteristics: irreplaceability, essential role in biotic cycles, scarcity, system-like properties, and its bulky nature, which complicates storage and trade (Savenije, 2002).

Climate change (CC) significantly disrupts atmospheric freshwater circulation, leading to altered precipitation patterns characterized by extreme events (Calvin et al., 2023;

Makarieva & Gorshkov, 2007; Sheil & Murdiyarso, 2009). These changes are expected to disproportionately reduce renewable water resources in dry subtropical regions, intensifying competition for water between human populations and ecosystems (Asadieh & Krakauer, 2015). Rural areas are likely to face severe freshwater scarcity, threatening food security and exacerbating social and economic tensions. Additionally, extreme rainfall events are projected to directly challenge water management and storage systems (McCornick, 2014).

Chile has been severely impacted by a megadrought. Between 1965 and 2019, four major drought events have been identified, underscoring the region's vulnerability to prolonged dry periods (Álvarez-Garretón et al., 2021). This large-scale phenomenon had profound consequences for ecosystems, the economy, and society. For instance, the reduction of glacial and snow cover has significantly limited streamflow during periods of high-water demand (World Bank, 2021). The OECD (2024) further emphasizes Chile's escalating water crisis in its 2024 Report, attributing it to a combination of high-water demand, erratic supply, and overallocation of resources. These factors illustrate a well-documented case of how CC exacerbates pressures on both human societies and ecosystems. The magnitude of the signal is evident in the record used here: mean annual precipitation across the studied basins fell from 1,053 mm in the 1980s to 739 mm over 2010–2020.

Compounding these challenges, water management in Chile is governed by the 1981 Water Law (Código de Aguas), which operates under a market-based model that inadequately addresses regional climatic variability. This regulatory shortfall has exacerbated social conflicts and intensified competition for water resources (Bauer, 2004a; Prieto, 2016). Furthermore, the rapid economic growth since the 1980s has placed additional strain on water resources. This has led to the overexploitation of aquifers (Fercovic et al., 2019; Aitken et al., 2016), raising concerns about the future stability of water markets in the face of CC.

The market-based design rests on a testable proposition. If tradable rights allocate water efficiently, then the price of a right should carry information about the physical availability of the resource it entitles its holder to use. Scarcity should be visible in prices, and prices should in turn discipline the use. This study asks whether that proposition holds in the Chilean case. Using water rights (WR) transactions from the Mediterranean zone, we estimate a dynamic spatial Durbin model that separates the influence of hydrological conditions from that of market dynamics, institutional signals and spillovers between basins, and we then ask what the estimated relationship implies under two contrasting climate futures. Three questions organize the analysis: whether WR prices respond to drought, at what temporal scale and spatial extent any such response operates, and whether the response is strong enough to matter relative to the market and administrative forces acting on the same prices. To the best of our knowledge this is the first assessment of the Chilean market that jointly identifies own-basin and neighboring-basin effects within a specification selected by test rather than assumed. Previous research has been implemented mostly at basin level and with limited incorporation of dynamics across time.

2. Background

2.1 Climate scenarios and water allocation

Human activities have caused global warming through the sustained release of greenhouse gases, producing shifts in the frequency and intensity of climatic extremes (Calvin et al., 2023). Global mean surface temperature is projected to exceed 1.5°C above pre-industrial levels under most scenarios by the end of the century, with the range across scenarios spanning 0.3°C to 4.8°C (Calvin et al., 2023). The spread across that range is not a measure of scientific uncertainty so much as of the socioeconomic trajectories the world may follow, which is why scenario analysis rather than single-path forecasting is the standard basis for assessing climate impacts.

Recent reports of The International Panel of Experts of Climate Change (IPCC) combine Representative Concentration Pathways, which describe trajectories of radiative forcing, with Shared Socioeconomic Pathways (Schipper et al., 2023). O'Neill et al. (2014) define SSPs as reference trajectories describing plausible societal development in the absence of climate policy. They are organized along two axes, the socioeconomic challenges to mitigation and those to adaptation, producing five narratives: SSP1 (sustainability) with low challenges on both axes, SSP2 an intermediate case, SSP3 (regional rivalry) high on both, SSP4 (inequality) low for mitigation and high for adaptation, and SSP5 (fossil-fueled development) high for mitigation and low for adaptation.

Applied to water, this framework is useful precisely because scarcity is not determined by supply alone. Water deficits depend on how scarce water is allocated among competing uses as much as on how much of it there is (Bijl et al., 2018), so scenarios that differ in socioeconomic development generate substantially different pictures of future water stress even under comparable hydrological change (Ndehedehe & Adeyeri, 2025). Coupling climate and socioeconomic pathways has accordingly become the standard basis for assessing future water availability and demand (Dau & Adeloye, 2021). That allocation dimension is here central as this study takes SSP1-2.6 and SSP5-8.5, which bracket the plausible range rather than representing a single best and worst case and does not strictly ask whether water will be available but what a market-based allocation mechanism does with the scarcity it faces.

2.2 Water scarcity and climate change

Accurately measuring the impact of CC is essential for designing effective mitigation strategies (Calvin et al., 2023). Since the late 1980s, progress has been made in public policies addressing freshwater scarcity (Booker et al., 2012). Loch (2013) demonstrated the profound interconnection between CC and freshwater management, emphasizing the need for policies and governance frameworks to align with this relationship. As future variations in water supply become increasingly likely, users at various governance levels will need to adapt, incurring associated costs. These challenges could be mitigated through the implementation of nature-based solutions (NBS) and by leveraging future opportunities for sustainable water management.

Efforts to address water scarcity have spurred innovative approaches that view water markets as potential opportunities (Grafton et al., 2012; Wheeler et al., 2017). However, Garrick et al. (2020) argue that conventional economic policy recommendations have fallen short, as they often overlook the complexities of water policies, and the distributive conflicts embedded in water allocation decisions. This is partly due to the failure to recognize

freshwater as an atypical market good. Furthermore, such approaches tend to neglect the ecological impacts of water management, focusing primarily on social and economic processes. In contrast, the adoption of NBS appears promising for enhancing freshwater availability, though their implementation remains limited and marginal to date (Vörösmarty et al., 2018).

In Chile, Barría et al. (2019) found that freshwater scarcity in the Perquilauquén River sub-basin is driven by a combination of CC effects and systemic issues such as over-allocation of WR, inadequate oversight, and the absence of mechanisms to adapt to mitigate climate variability. These factors are projected to further intensify water scarcity in the future. Similarly, Muñoz et al. (2020) concludes that the water crisis in Petorca River basin is primarily driven by CC and unsustainable water management practices. Compounding the problem, excessive WR allocation amounting to 18% of the basin's mean annual precipitation, have significantly worsened the crisis. Weak regulatory frameworks and overexploitation have resulted in severe water shortages, leaving communities at risk of losing access to drinking water.

2.3 The value of water

Economic theory posits that the efficient use of water as a resource requires pricing it at its long-term marginal cost, which reflects the additional benefit derived from its use in various economic activities. Such pricing would theoretically promote optimal resource allocation. During droughts, when demand exceeds supply, prices should rise to balance the two. However, this approach faces practical challenges, particularly due to environmental externalities that are often uncertain and difficult to quantify (Grafton et al., 2020; Schulz et al., 2024). Furthermore, issues such as WR allocation, governance inefficiencies, and social equity concerns complicate the implementation of a purely market-based solution. Despite these obstacles, global freshwater use has continued to rise at an annual rate of approximately 1% since the 1980s (UN, 2021), underscoring the urgent need for improving water management policies that reconcile economic efficiency with sustainability. Muñoz et al. (2020) highlight that a key aspect of water resource management is the reuse and proper management.

The prediction that drought raises the price of a water right holds only if the right reliably delivers water. Where entitlements are nominal and their exercise depends on physical availability, drought reduces the expected quantity of water attached to a right and may therefore reduce rather than raise its value. Which of these two forces dominates is an empirical question, and the answer characterizes whether a market transmits scarcity or merely records it.

WR have been advocated as an efficient mechanism for resource allocation, yet they face criticism for their multifaceted impacts (Grafton et al., 2016). Transaction costs, for instance, often impede effective and efficient exchanges under existing legal framework (Leonard et al., 2019; Loch et al., 2018). In Australia, Loch (2013) demonstrated that trading water allocations serves as a critical risk management tool during periods of scarcity, particularly during severe droughts. Grafton et al. (2016) further argue that water markets, when combined with effective planning and regulation, can yield significant economic,

environmental, and social benefits. Similarly, Arellano-Gonzalez et al. (2021) highlight that in California, more efficient allocation of water reductions could mitigate the costs of water scarcity, suggesting that institutional and market reforms present valuable opportunities for adaptation in dynamic and uncertain scenarios. Wang et al. (2025) compare the system implemented in China with the Chilean framework, showing that China's legal architecture remains incomplete and, despite strong government involvement, conflicts and tensions among water-market users persist.

2.4 The Chilean case

The Chilean market-based model to manage freshwater scarcity, regulate misuse by separating WR from land ownership (Prieto et al., 2019), and establishes a uniform management formula at national level, declaring water as a national good for public use while guaranteeing perpetual usage rights, granted free of charge, to those who register them. The General Directorate of Water (DGA) was granted authority to register WR, by assigning perpetual and freely transferable water usage rights on a first-come, first-served basis. Once assigned, owners may freely trade these WR (Donoso, 2018). The market was designed to reallocate water to its highest economic value by limiting demand through the transfer of rights and market forces, ensuring minimal public intervention and relying heavily on water user associations (Donoso, 2018). Evidence suggests that this management scheme has made WR more competitive where water is scarce (Herrera et al., 2019), but this uniform application across Chile happens without considering significant hydrological and productive differences between basins (Bauer, 2004a; Prieto, 2016).

Moreover, the original 1981 framework did not require rights holders to demonstrate actual, immediate use of the water they claimed, speculative entities accumulated vast portfolios of WR at zero initial cost, holding them merely to extract monopoly rents or to secure future industrial expansion. The resulting socio-economic landscape heavily favored agro-export conglomerates and transnational mining corporations, creating a deeply asymmetric playing field (Roose & Panez, 2020; Perez-Silva & Castillo, 2023). Additionally, the lack of hydrological information generated long-term overallocation of WR (Barría et al., 2019), with estimations around 30% of the national catchments (Reade Malagueño & D'Odorico, 2024).

The DGA is tasked with assisting in conflict resolution among water users, but without a dedicated budget (World Bank, 2011) which undermines its ability, leading to market distortions (Cristi et al., 2013). In 2005 the water law was reformed to address these challenges by introducing a non-use fee, though its effectiveness remains uncertain (Valenzuela et al., 2013). Subsequent regulations, including requirements for intended water use, the establishment of ecological flow rates (Bauer, 2015; Donoso, 2018), and public bidding in cases of high-water demand, have also proven largely ineffective (Peña-Torres et al., 2019). This is partly because WR issued prior to the reform remain unaltered, limiting the impact of these measures (Bauer, 2015). This has led to the overexploitation of aquifers and necessitated adjustments in administrative policies (Aitken et al., 2016; Fuster et al., 2024; Hernandez & Lobos, 2024).

Alongside the allocation regime, the DGA may declare a zone under water scarcity (*decreto de escasez hídrica*, SD hereinafter), a temporary administrative measure that acknowledges a deficit (DGA, 2025) and enables extraordinary management powers. These declarations are public, discrete and dated, which makes them an observable institutional signal of scarcity that is distinct from, though correlated with, the underlying hydrological conditions. The analysis below estimates the effect of each separately.

While there's evidence of progress in regulating Chile's water markets, significant challenges remain (Cristi et al., 2013). Transactions are conducted through private negotiations, where prices are determined by the information and bargaining power of the involved parties (Cristi et al., 2013; Donoso, 2018). Evidence indicates that water scarcity is driven by poor market and resource management, deregulation, a lack of territorial planning and integrated resource management, short-term decision-making, and widespread distrust among stakeholders (Babidge, 2016; Urquiza & Billi, 2020; Dardati & Sabat, 2024).

The framework was substantially amended by Law N° 21,435 (2022), which changed newly granted rights into renewable administrative concessions, prioritized human consumption, strengthened non-use forfeiture and DGA intervention, and restricted rights over glaciers and protected wetlands. Since the transactions analyzed predate the reform, our results characterize the former regime and should be interpreted as a diagnosis of the system that the reform sought to address.

3. Methods

3.1 Characterization of transactions

The study utilizes the mean transaction price of WR within river basins, derived from a comprehensive database of the Superintendence of Sanitary Services (SISS). This database encompasses over 15,000 WR transactions recorded nationwide between 1993 and 2014, and data are used to utilities tariffication processes. The subsample of recorded prices is used to estimate the value of raw water at basin level. Each transaction was geographically characterized by integrating it with relevant climatic and socioeconomic data. The transactions are documented at the municipal level and linked to specific water sources, allowing for accurate assignment to the corresponding river basins by the DGA.

The study focused on a selection of Chilean river basins characterized by a short range of climatic conditions and hydrological connectivity, enabling the analysis of climatic and spatial spillover effects across river basins. Basins were chosen based on the criterion of having recorded more than 30 WR transactions between 2005 and 2014. To consider direct hydrological connectivity, only basins situated between latitudes 30°S and 42°S were included in the analysis. Two basins meeting these criteria were subsequently excluded because the catchment precipitation record required to construct the drought index was unavailable for them. This selection process yielded 8,020 individual transactions across fifteen river basins, organized as a monthly panel of 1,785 basin-month observations spanning 2005–2014 (Figure 1).

The dataset revealed extended periods with no recorded transactions in most basins, and 60.4% of basin-months contain no transaction at all. Where a month records no transaction,

the price series carries the last observed value forward, so that a basin may hold an unchanged price for a long interval. The median basin holds a constant price for 49 consecutive months at its longest run. This is a property of the price series itself rather than an analytical choice, since a price can only be observed when a transaction occurs, but it has consequences for the estimated persistence of prices that must be confronted directly. We therefore report the persistence coefficient on three progressively more demanding subsamples in Section 4, and we treat the full panel estimate as an upper bound.

3.2 Construction of the drought index

Hydrological conditions were characterized using the Standardized Precipitation Index (SPI), which expresses accumulated precipitation as a probability transform onto the standard normal distribution (Alvarez-Garreton et al., 2018; Koudahe et al., 2017). Catchment-level monthly precipitation for the fifteen basins was obtained for 1979–2020, a forty-two-year record that satisfies the length requirement for a stable distributional fit and comfortably exceeds the World Meteorological Organization’s thirty-year minimum. The index was computed by accumulating precipitation over a window of k months, fitting a two-parameter gamma distribution to the strictly positive accumulations of each calendar month separately, and applying Thom’s mixed-distribution correction, $H(x) = q + (1 - q)G(x)$, where q is the empirical probability of zero precipitation. Fitting each calendar month separately is necessary in a strongly seasonal Mediterranean climate, and the zero correction is necessary because dry-season months frequently record no precipitation at all, in some basins in more than a fifth of peak summer months. Three accumulation windows were computed, at three, six and twelve months, and all three were carried through estimation. All are reported because the accumulation scale at which a market responds to drought is itself a result, and restricting attention to a single window would conceal it.

3.3 Development of the explanatory model

The dynamic Spatial Durbin Model (SDM) was employed to analyze the data. This model incorporates both temporal and spatial dimensions (Anselin, 1988; LeSage & Pace, 2009). The dataset was organized as a panel data within a matrix structure which allows us the explicit consideration of spatial dependencies and temporal dynamics, providing a robust basis for evaluating spillover effects across river basins (Elhorst, 2014). The SDM has recently been used to separate local from spillover effects in water resource systems, including regional supply suitability, watershed water security and water–energy–food efficiency (He et al., 2024; Zhang et al., 2024a, 2024b; Liu et al., 2025; Guo et al., 2026).

Spatial relationships between basins were represented by a row-standardized weight matrix $\mathbf{W}$ built from the four nearest neighboring basins, measured between basin centroids. The number of neighbors was selected primarily due to the limited number of basins retained after the initial filtering process. Because the panel is observed repeatedly over time, $\mathbf{W}$ is defined once at basin level and blocked across periods, so that each link connects distinct basins within the same month and no link connects a basin to itself in another month.

The estimated specification is:

$$y_t = \rho \mathbf{W} y_t + \tau y_{t-1} + \theta_\tau \mathbf{W} y_{t-1} + X_t \beta + \mathbf{W} X_t \theta + \mu + \iota_N \xi_t + \varepsilon_t \quad (1)$$

where y_t is the $N \times 1$ vector of log mean transaction prices across the fifteen basins in month t , $\mathbf{W}$ the row-standardized spatial weight matrix, ρ the spatial autoregressive parameter, τ the coefficient on the temporally lagged dependent variable, θ_τ the coefficient on its spatial lag, β the vector of coefficients on the basin-level covariates X_t , θ the vector of coefficients on the spatially lagged covariates $\mathbf{W}X_t$, μ a vector of basin effects, ξ_t a period-specific effect, and ε_t the idiosyncratic error. Estimation is by maximum likelihood.

Equation (1) nests simple spatial forms and allows to set $\theta = 0$ and $\theta_\tau = 0$ to yield a spatial autoregressive model in which the ratio of indirect to direct effects collapses to $\rho/(1 - \rho)$ and is therefore identical for every covariate. The Durbin terms were retained or discarded based on a likelihood ratio test against that restriction rather than assumed. The test was repeated for each accumulation window of the drought index.

Effects were decomposed following LeSage and Pace (2009). For covariate k the multiplier matrix is $S_k(\mathbf{W}) = (\mathbf{I} - \rho\mathbf{W})^{-1}(\beta_k\mathbf{I} + \theta_k\mathbf{W})$, whose mean diagonal gives the direct effect, whose mean row sum gives the total effect, and whose difference gives the indirect effect. Reporting the decomposition per covariate rather than as a single aggregate is essential, since the covariates are measured in incommensurable units and an average across them has no interpretation.

3.4 Variables

Table 1 reports the variables entering the model with their descriptive statistics over the estimation sample. The variables were constructed from the same transaction records and from public administrative sources. These comprise the number of transactions per basin-month, the mean flow transacted in liters per second, an indicator of mining presence in the basin, and an indicator for a SD in force in the previous month, together with the interaction between transacted flow and the SD. Prices are reflected in Chilean Pesos, but adjusted by UF (*Unidad de fomento*): UF is a inflation-indexed unit of account used in Chile to preserve the real value of financial obligations by adjusting daily to variations in the Consumer Price Index (Banco Central de Chile, 1989, art. 35, num. 9), to avoid the overdispersion of raw prices within and across basins. Mining presence is a binary indicator created by the relative weight of the mining activities in the basins, based on the percentage of income from mining patents, with a threshold of more than 2.5% of municipal GDP as suggested by Oyarzo and Paredes (2021).

SPI-12 averages -0.090 across the estimation window, so 2005–2014 was drier than the 1979–2020 climatology against which it is scored. This is consistent with the onset of the megadrought, and it sets the baseline from which the scenario projections in Section 4.3 depart.

3.5 Scenario evaluation

To assess the implications of projected CC for WR prices, the estimated model was evaluated under the SSP1-2.6 and SSP5-8.5 scenarios for four future periods: 2021–2040, 2041–2060, 2061–2080 and 2081–2100. Projected precipitation was obtained from the MIROC6 general circulation model via the IPCC WGI Atlas (Iturbide et al., 2022). Because SPI-12 is an annual accumulation and the projections are supplied as annual precipitation totals, the projected

and estimated indices are the same quantity and were produced by the same gamma transform against the same 1979–2020 basin climatology. Socioeconomic variables were held at their basin means over the estimation period, as there is insufficient basis for projecting changes in these variables.

Two corrections were applied. First, both scenarios sit approximately 32% below the observed climatology already in the first projected period, and they agree with one another there to within 1.7%. Since the scenarios have not yet diverged at that date, the shortfall identifies model dry bias rather than a climate signal and scoring raw model output against an observed-precipitation gamma would register that bias as permanent drought. Projections were therefore expressed as fractional change relative to the first projected period and applied to the observed basin climatology, following the standard delta-change approach. Second, prices were obtained as the long-run equilibrium of equation (1) rather than as a single-period linear prediction. Imposing $y_t = y_{t-1}$ gives:

$$y^* = [(1 - \tau)\mathbf{I} - (\rho + \theta_\tau)\mathbf{W}]^{-1} (\alpha + \mathbf{Z}) \quad (2)$$

where $\mathbf{Z}$ collects every covariate other than the lagged price together with its spatial lag. Uncertainty was propagated by drawing 4,000 parameter vectors from the estimated coefficient distribution and solving (2) for each, discarding any draw for which no stable steady state exists. Results are reported as medians with 90% intervals.

4. Results and Discussion

Table 2 reports the estimated model in both its autoregressive and the Durbin form. The likelihood ratio test rejects the autoregressive restriction decisively (LR = 33.35 on 7 degrees of freedom, $p < 0.0001$), and the Akaike criterion agrees, so the Durbin specification is retained, and interpretation proceeds from it. Under the autoregressive restriction the ratio of indirect to direct effects is forced to take the same value for every covariate, which would have concealed the sign reversal reported below.

Prices respond to accumulated precipitation, but only at the annual scale (Figure 2). The three- and six-month accumulation windows produce coefficients indistinguishable from zero in both specifications, while the twelve-month window is significant in both (0.102, $z = 2.34$ in the autoregressive form and 0.290, $z = 2.85$ in the Durbin form). The opposition between own-basin and neighboring-basin responses discussed below likewise appears only at the annual window. A WR is a durable asset whose value should track multi-season availability rather than recent weather, and the estimated pattern is consistent with that reading. It also indicates that studies relying on short-window drought indices may report null results that reflect the choice of window rather than the absence of a relationship.

The own-basin effect is positive. A one-unit improvement in SPI-12 raises prices in the basin where it occurs by roughly a third, so drought lowers them, which inverts the intuition that scarcity raises price. Section 2.3 anticipated this outcome: where an entitlement is nominal and its exercise depends on physical availability, drought reduces the expected quantity of water attached to the right, and the asset devalues accordingly. The market records scarcity rather than transmitting a scarcity premium.

Administrative signals act far more strongly than hydrological ones. A SD in force in the previous month raised prices by about half (0.426, $z = 4.74$), an effect five times (*i.e.*, 0.475 vs. 0.087) the magnitude of a one-unit change in the drought index and far more precisely estimated. The interaction between transacted flow and the decree is negative and highly significant (-0.025 , $z = -5.42$), indicating that larger transactions attract proportionally lower unit prices under declared scarcity. Market variables behave as expected, with the number of transactions raising prices (0.009, $z = 4.14$) and larger mean flows per transaction lowering them (-0.0009 , $z = -3.99$).

Contemporaneous price spillovers are absent. The spatial autoregressive parameter is not significantly different from zero in the Durbin specification ($\rho = -0.032$, $z = -0.86$), while the spatial lag of the previous month's price is significant and positive (0.112, $z = 2.68$). Prices therefore transmit between basins with a delay rather than simultaneously, which describes a diffusion process across space and time rather than a spatial equilibrium. The apparent contemporaneous dependence in the autoregressive specification ($\rho = 0.079$, $z = 3.52$) does not survive the inclusion of the Durbin terms, which is a further reason to prefer the general specification.

4.1 Direct, indirect and total effects

Figure 3 reports the effects of decomposition. Four covariates have spillover effects distinguishable from zero, and in two of these the spillover opposes the own-basin effect in sign. Neither pattern can arise under the autoregressive restriction, which forces the ratio of indirect to direct effects to equal $\rho/(1 - \rho)$ for every covariate alike, here 0.086. The observed ratios instead range from -1.05 to 8.46 .

The drought index carries a positive direct effect of 0.291 and a negative indirect effect of -0.203 , so wetter conditions in a basin raise its prices while wetter conditions among its neighbors lower them. Both intervals exclude zero, which deserves comment because the spatially lagged coefficient from which the indirect effect derives does not reach conventional significance on its own (Table 2). Inference on effects and inference on coefficients need not agree, since the effect combines ρ and θ and the two covary, and it is the effect rather than the coefficient that carries the economic interpretation. The asymmetry is readily interpreted as substitution: when water is physically available nearby, buyers hold alternatives, and local entitlements lose scarcity value. It also implies that regionally synchronous drying, which is what climate projections describe, produces a far weaker aggregate response than a localized drought would, since the two channels largely offset one another. The total effect of 0.087 is an order of magnitude smaller than the direct effect.

Mining presence shows the opposite configuration. Its direct effect of 0.055 cannot be distinguished from zero, while its indirect effect of 0.480 is estimated with an interval well clear of it, so extractive activity in adjacent basins moves prices where extractive activity within the basin does not. Demand pressure from mining is evidently transmitted through the market rather than confined to the basin where the activity is located, which bears on the recurring proposal to manage water at basin scale in isolation.

The number of transactions has direct and indirect effects that almost exactly cancel, at 0.010 and -0.010, and both intervals exclude zero. The offsetting is therefore a property of the estimates rather than a consequence of imprecision. Market activity redistributes value between basins without generating any at the regional scale, a result invisible in any specification that constrains the two effects to share a sign.

The SD behaves differently again. Its direct effect is large and precisely estimated, while its spillover cannot be distinguished from zero, so an administrative declaration moves prices in the basin it names and leaves neighboring basins largely unaffected. The market reads the decree as information about a particular basin rather than as a signal about regional conditions.

4.2 Robustness of the persistence estimate

The lagged price carries the largest coefficient in the model, and because 60.4% of basin-months record no transaction the price series is partly a carried-forward one. Persistence estimated on such a series is mechanically inflated. Regressing price on its own lag gives 0.823 on the full panel, 0.715 when restricted to months in which a transaction was observed, and 0.582 when restricted to consecutive observed months. Persistence is therefore real but overstated by the structure of the record, and the full-panel figure should be read as an upper bound. The substantive results are unaffected, since the drought, decree and spillover coefficients are estimated from variation across basins and months rather than from the persistence channel.

Including maximum monthly precipitation alongside SPI-12 leaves the drought result intact and slightly strengthens it (0.330, $z = 3.15$), while maximum precipitation itself is not significant in the Durbin specification. The two are alternative measures of the same underlying phenomenon, and the accumulated index is the theoretically appropriate one, so the reported specification retains SPI-12 alone.

4.3 Model evaluation in the climate change context

Under SSP1-2.6, projected precipitation is essentially unchanged across the century and equilibrium prices are correspondingly flat, varying by less than two percent in any period. Under SSP5-8.5, precipitation declines by roughly a quarter by 2081–2100 and prices fall by 22.8% (Figure 4). The direction is robust, with 97% of draws yielding a decline, but the magnitude is uncertain by a factor of nine, the 90% interval for the final period running from -39.6% to -4.3%.

The projected decline is smaller than the underlying precipitation change alone would imply, and it runs downward. The first follows from the substitution effect identified in Section 4.1. Climate projections describe regionally coherent drying, and under such drying the positive own-basin effect and the negative neighboring-basin effect largely cancel, so a market that would respond sharply to a localized drought responds weakly to a general one. The second is more consequential. Under the most severe scenario considered, WR become less valuable as water becomes scarcer, which is the opposite of what a functioning scarcity-pricing mechanism would produce.

While the model has limitations in projecting prices, it serves as an indicator of projected scarcity under the most severe CC scenarios. According to Valdés-Pineda et al. (2014), a critical issue in Chile is that basins with the highest water demand are those with the largest populations and significant mining activity, primarily located in central and northern Chile. This has sparked discussions about exporting water from southern basins. While such measures could alleviate resource pressure, they fail to account for other critical factors, such as ecosystem services related to regulation and provision, as well as impacts on ecosystems, biodiversity, species invasions, and system stability. Our finding that mining presence acts primarily through neighboring basins rather than within them adds a further consideration to that debate.

In addition to unaddressed ecological effects, governance challenges arise, as changes in projected precipitation may affect WR prices through both local market conditions and spatial interactions among neighboring basins. Traditional solutions are clearly insufficient to address these complex issues; instead, innovation and a focus on the root problem—water use rather than mere provision or scarcity—are required (Lopez-Gunn & Llamas, 2008).

4.4 Policy implications and limitations

The general lack of information on WR transactions introduces biases for model development. Previous national-level analyses have shown that socioeconomic characteristics of a basin can influence WR prices (Araos & Roco, 2025). This study estimates the effects of drought and spatial spillovers between basins, revealing that neither operates in the direction the market-based framework assumes. This underscores the need for policies and territorial planning to adopt a holistic, system-wide perspective, as changes in one basin may affect others through market and spatial dynamics.

Most studies focus on determinants of WR price transactions without considering spatial variation or inter-basin effects. This study provides evidence of the influence of neighboring basins and climatic factors, demonstrating that homogeneous formulas are inadequate.

Currently, the minimum ecological flow required to maintain WR, as mandated by the government, corresponds to 20% of the river basin's mean flow (MMA, 2012). This policy applies to WR established after that date, while maintaining minimum flows for rights granted earlier. However, this policy overlooks significant ecological differences among basins, which have evolved over time (e.g., during drought periods), and fails to account for varying minimum flow requirements needed to sustain ecosystems. Legal and institutional frameworks play a critical role in shaping transaction costs and market functionality. Evidence from Colorado indicates that legal uncertainty and third-party injury rules can substantially raise transaction costs, even in active water markets (Womble & Hanemann, 2020). Chile's water market may be transitioning into a phase like the 'fifth stage' of water management, as proposed by Loch et al., characterized by heightened uncertainty and the need for adaptive governance mechanisms in response to socio-environmental pressures (Loch et al., 2020).

SD constitute an important state intervention for managing water during periods of severe hydrological stress. In Chile, their relevance is heightened by persistent drought and spatially uneven water availability, as well as by structural over-allocation and localized

overexploitation of water resources (Garreaud et al., 2017, 2020, 2025; Barría et al., 2019; Taucare et al., 2024). In this context, SD can influence the effective allocation and redistribution of water among competing users, with potentially significant socioeconomic and spatial implications within Chile's market-based water governance system (Bauer, 1998, 2004a, 2004b; Donoso, 2015). The strength of the SD effect relative to the drought effect carries a policy reading of its own. The market responds far more sharply to an administrative declaration of scarcity than to the hydrological conditions that declaration describes. Where this reflects the informational value of a public signal in a market conducted through private negotiation, it suggests that improving the timeliness and granularity of hydrological information could substantially improve price formation. Where instead it reflects the regulatory consequences that follow a declaration, it suggests that the market is pricing administrative risk rather than water. Distinguishing the two is beyond what the present data allow, but the distinction is consequential for the design of the concession regime introduced by Law 21,435.

Environmental flows represent the specific flow regime necessary to maintain a river's aquatic habitats and ecosystem processes (UN, 2021). Identifying the ecosystem services provided by a basin and valuing them allows the determination of a minimum hydrological regime required to deliver those services. This minimum becomes the flow maintenance value for the basin, and recovery actions are necessary for basins that have fallen below this threshold. NBS can be applied in such cases to reduce operational and capital costs related to water or to enhance productivity in drought-affected basins (Abell et al., 2017; Garrick et al., 2017; Horne et al., 2017). NBS, combined with water capture management, can increase groundwater recharge, conserve surface water, enhance soil moisture retention, and improve regional precipitation management (UN, 2021).

The past and forthcoming reforms of Chile's Water Code are highly relevant for understanding the evolution of water governance and the functioning of water markets. The 2005 reform sought to address speculative WR accumulation through the introduction of non-use fee, while the 2022 reform strengthened state intervention by introducing stricter conditions on the use and potential expiration of WR (Donoso, 2015; Perez-Silva & Castillo, 2023; Adiego et al., 2025). These reforms, together with forthcoming regulatory changes, are particularly important in a context of persistent water scarcity and structural over-allocation, as they may significantly affect the incentives to hold, trade, and use WR and, consequently, the spatial and socioeconomic distribution of water resources.

Public policies must be adjusted to reflect the specific characteristics of each river basin, with NBS serving as a viable alternative to maintain ecological flows in each unit (Arellano-Gonzalez et al., 2021). Critiques of the WR transaction system highlight its focus on water use rather than the preservation of water sources. However, similar studies emphasize that regulating water use types and improving efficiency should be prioritized (Valdés-Pineda et al., 2014). In markets with structural similarities to the Chilean case, such as Australia, the composition of agricultural activity—particularly the prevalence of permanent crops—has been shown to influence the direction of water allocation flows under scarcity conditions (Zhao et al., 2024). In the absence of formal regulatory frameworks, cooperative mechanisms among agricultural users—such as those observed in northern Chile—can emerge as effective alternatives. These cooperatives implement static redistribution rules to manage

allocation uncertainty, a phenomenon also modeled in the semi-arid north of Chile (Delorit & Block, 2018).

This approach does not contradict the principles of water use and markets but rather enables sustainable management of this scarce and irreplaceable resource. Traditionally, the economic value of water sources has been determined by use and extraction costs. However, this narrow perspective limits analysis and overlooks potential efficiency measures that could be implemented post-extraction. According to Abell et al. (2017), Garrick et al. (2017), and Horne et al. (2017), investment in river basin conservation could yield significant returns.

McNamara et al. (2021) emphasize that the hydrological responses to CC, land-use pressures, and irrigation expansion vary substantially across basins, underscoring that effective adaptation requires basin-specific planning rather than uniform national prescriptions. This can be addressed by the development of NBS as an alternative for recharge and the establishment of mechanisms to improve water use efficiency and control. These measures are essential for ensuring the long-term sustainability of water resources in Chile.

The management of water in Chile reflects a critical decoupling between biophysical dynamics and its institutional framework. By treating WR as commodities decoupled from land and ecosystem boundaries, the legal architecture ignores the continuous, interconnected nature of river basins. This rigid administrative approach fails to adapt to climate-driven hydrological changes—such as severe droughts and shifting snowpacks—creating a structural disconnect where static legal entitlements conflict directly with the dynamic ecological realities of water scarcity.

Several limitations qualify these results. The transaction data do not distinguish among right types with systematically different prices, while the SISS price denominator cannot be fully verified. The 60% of basin-months without transactions limits inference on short-run dynamics, and the small number of basins reduces the reliability of spatial inference. The scenario exercise relies on a single climate model, fixed socioeconomic conditions, and a conservative delta correction. Most importantly, it reflects the 2005–2014 regulatory regime and therefore should not be interpreted as a forecast under the current regime following Law 21,435. Additional variables such as land use, industrial activity, and water use could improve robustness, but their limited temporal and spatial resolution would require interpolation and potentially introduce bias.

5. Conclusions

This study asked whether the Chilean water market prices scarcity, as the 1981 Water Code assumed it would. The answer is that it does so weakly, at only one temporal scale, and in the wrong direction. Prices respond to accumulated precipitation over twelve months but not over three or six, and the response indicates that drought lowers rather than raises the value of a water right. A market designed to allocate water efficiently under scarcity devalues the entitlement precisely when the resource becomes scarcer and therefore transmits no incentive to conserve it.

The spatial structure reinforces this reading. Wetter conditions in neighboring basins depress local prices by nearly as much as wetter conditions locally raise them, so that the regionally coherent drying described by climate projections produces a much weaker aggregate response than a localized drought would. Contemporaneous price spillovers between basins are absent, though prices do transmit with a lag, and demand pressure from mining acts principally through neighboring basins rather than within the basin where the activity occurs. Homogeneous, basin-by-basin regulation is poorly matched to a market that behaves this way.

Against these hydrological effects stands a much stronger institutional one. An administrative declaration of water scarcity raises prices by about half within a month, several times the magnitude of any climate variable in the model. The market responds to the announcement of scarcity far more than to scarcity itself. Whether this reflects the informational value of a public signal in a market conducted through private negotiation, or the pricing of regulatory risk, it indicates that price formation in this market is governed by institutional events rather than by physical conditions.

Scenario projections carry these estimates forward. Under aggressive mitigation, equilibrium prices are essentially unchanged across the century. Under high emissions, they fall by roughly a quarter by 2100, with the direction robust and the magnitude uncertain. These projections describe the regime that Law 21,435 was written to reform, and they support the direction of that reform while suggesting it does not go far enough. Converting perpetual rights into renewable concessions addresses the allocation of entitlements, but it does not by itself repair a price signal that moves the wrong way. Achieving that would require attaching value to the maintenance of the resource rather than to the entitlement alone, through environmental flow requirements that reflect basin-specific ecological conditions, through greater transparency in transaction records, and through recognition that basins are connected by a market even where they are not connected by a river.

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Data and code availability

The analysis pipeline, comprising the construction of the drought index, the spatial weight matrix, model estimation, the effects decomposition and the scenario projection, is available as documented Python scripts here: <https://github.com/horaciosamaniego/chile-water-rights-scarcity/>. Data are available on request to authors.

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Figure 1. Water right transactions. **A)** Total water right transactions per year between 2005 and 2014. **B)** Total water right transactions per river basin. Colors show the location of selected river basins in the map.

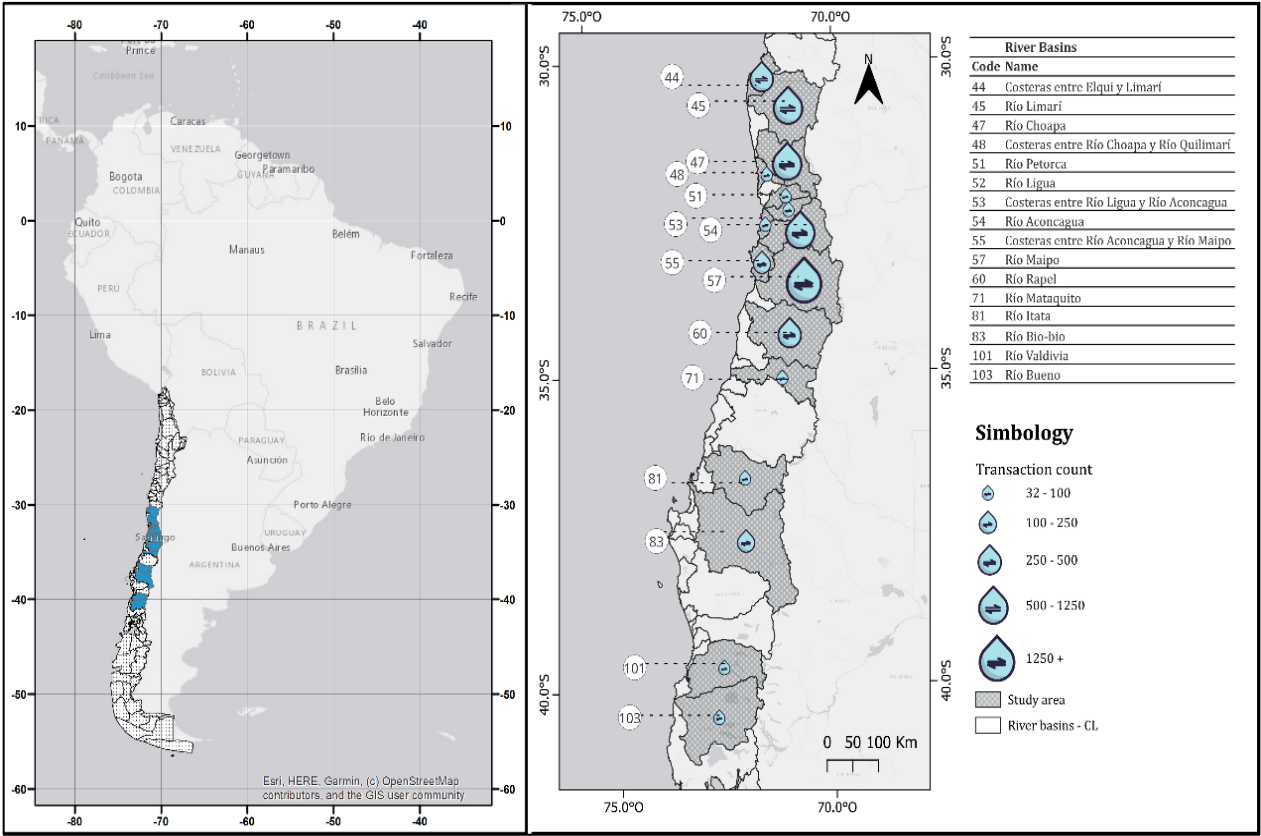


Figure 2. Response of water rights prices to drought by precipitation accumulation window. Own-basin and neighboring-basin coefficients on the Standardized Precipitation Index at three, six and twelve-month accumulation, estimated from the spatial Durbin specification, with 95% confidence intervals. Prices respond only at the annual scale, and the own-basin and neighboring-basin responses take opposite signs at that scale alone.

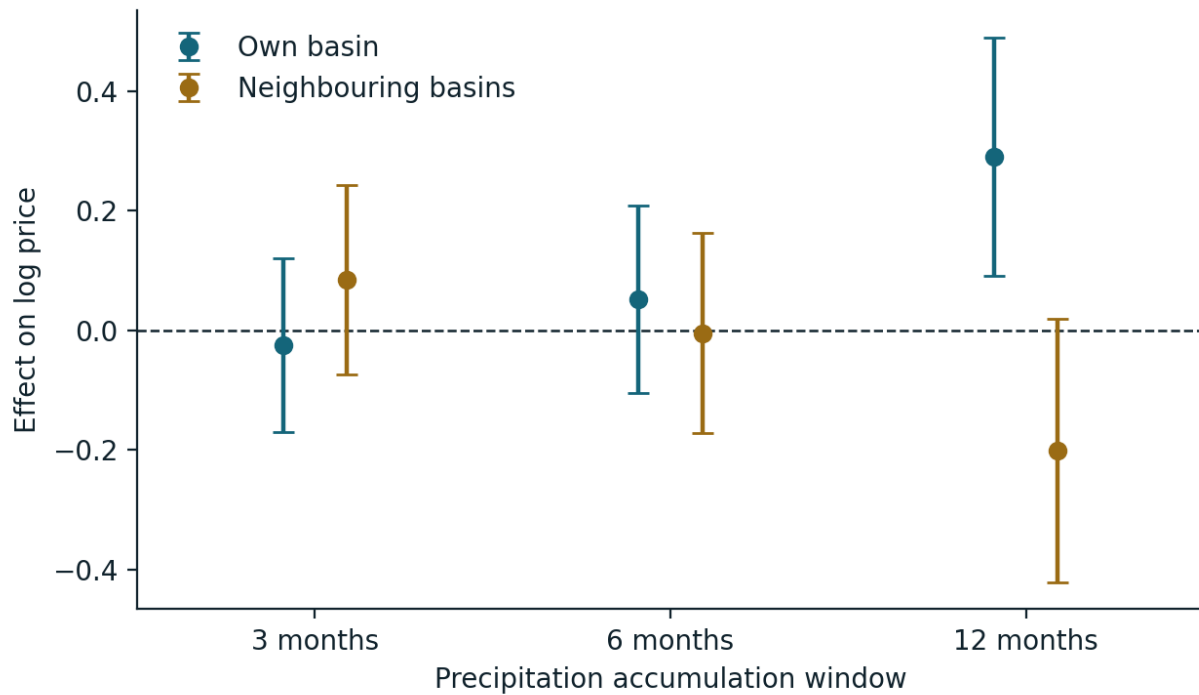


Figure 3. Direct and indirect effects of each covariate on water rights transaction prices. **(A)** Covariates whose effects are resolvable on the full scale. **(B)** The remaining covariates on a horizontal axis magnified roughly eightfold, so marker positions are not comparable between panels. Effects were computed from the spatial Durbin estimates in Table 2 following LeSage and Pace (2009), with medians and 90% intervals from 4,000 draws of the estimated coefficient distribution. Direct effects operate within the basin and indirect effects through neighboring basins. Filled markers denote intervals excluding zero, open markers intervals spanning it. Under the autoregressive restriction the two effects would be constrained to share a sign in every row, at a fixed ratio of $\rho/(1 - \rho) = 0.086$.

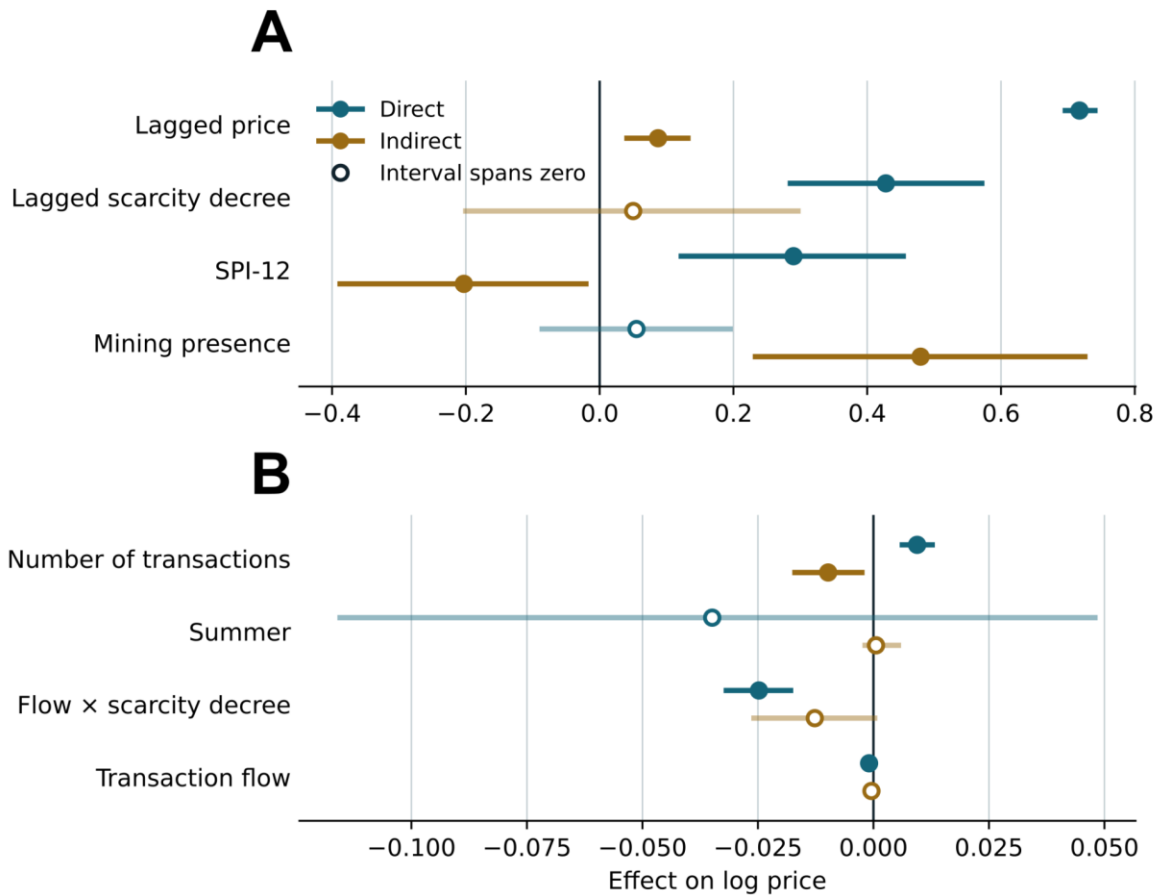


Figure 4. Projected change in long-run equilibrium water rights prices relative to the first projected period, under SSP1-2.6 and SSP5-8.5. Shaded bands show 90% intervals from 4,000 draws of the estimated coefficient distribution.

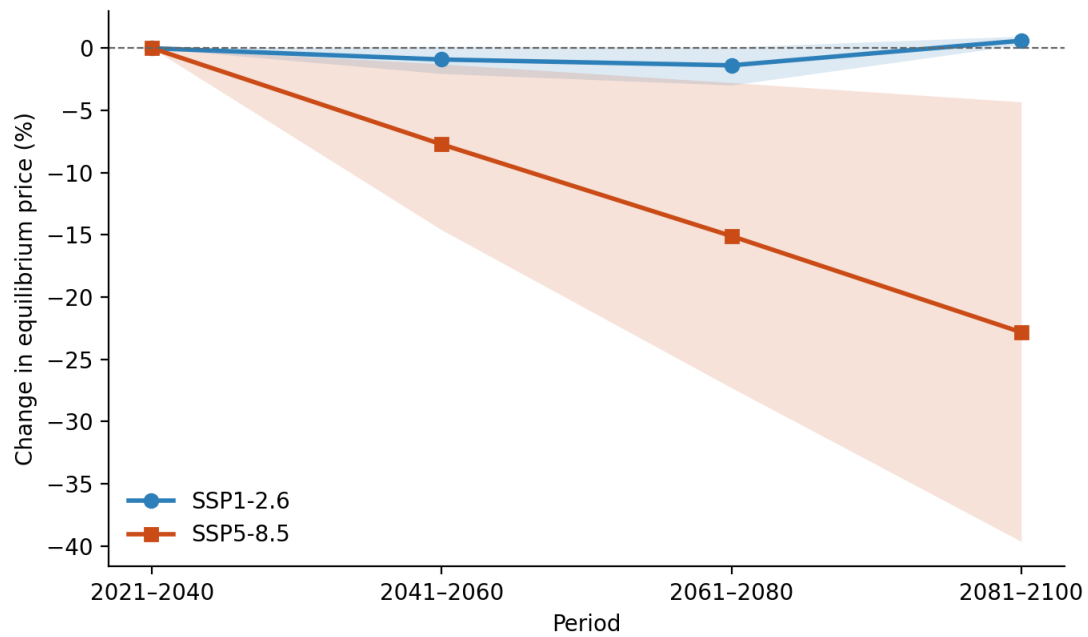


Table 1. Variables entering the model, with descriptive statistics over the estimation sample. Fifteen river basins of Mediterranean Chile observed monthly over 2005–2014, giving $N = 1,785$ basin-months after the loss of the first period to the lagged price. Prices are logarithms of the mean transaction price in UF. SPI-12 is the twelve-month Standardized Precipitation Index computed from catchment precipitation over 1979–2020, so its negative sample mean indicates that the estimation decade was drier than the reference climatology. Indicator variables take the value 0 or 1, and their means give the share of basin-months in the relevant state.

Variable	Description	Mean	Standard Deviation	Range
<i>Price</i>	Logarithm of mean transaction price (UF)	4.379	1.907	–2.98 to 12.82
<i>Lagged price</i>	Price in the previous month	4.370	1.904	–2.98 to 12.82
<i>Transaction flow</i>	Mean flow transacted per basin-month (l/s)	21.714	107.140	0.01 to 1400.0
<i>Number of transactions</i>	Transactions per basin-month	4.493	12.401	0 to 88
<i>Mining presence</i>	Indicator of mining activity in the basin	0.158	0.365	0 or 1
<i>Lagged SD</i>	Water scarcity decree in force, previous month	0.159	0.365	0 or 1
<i>Flow × SD</i>	Interaction of transacted flow and decree	0.917	5.671	0 to 129.2
<i>Summer</i>	Indicator for the austral summer half-year	0.496	0.500	0 or 1
<i>SPI-12</i>	Twelve-month Standardized Precipitation Index	–0.090	0.604	–2.03 to 1.71

Table 2. Determinants of water rights transaction prices in fifteen river basins of Mediterranean Chile, 2005–2014. Maximum likelihood estimates of eq. (1). The dependent variable is the logarithm of the mean transaction price in UF. Own-basin terms appear in the upper panel and spatially lagged terms in the lower. The SAR column imposes $\theta = 0$ and $\theta \sim \tau \sim 0$, so no spatially lagged covariates enter and dashes mark terms that specification excludes; the SDM column is unrestricted. $\mathbf{W}$ is row-standardized over the four nearest neighboring basin centroids and blocked across periods, so every link connects distinct basins within the same month. $\mathbf{W} \times \text{price}$ is the spatial autoregressive parameter ρ . A likelihood ratio test rejects the SAR restriction (LR = 33.35, df = 7, $p < 0.001$), and interpretation proceeds from the SDM column.

Variable	SAR				SDM			
	est.	s.e.	z	p	est.	s.e.	z	p
<i>Constant</i>	0.719	0.106	6.81	<0.001	0.792	0.138	5.73	<0.001
<i>Transaction flow</i>	-0.001	0.0001	-3.81	<0.001	-0.001	0.0001	-3.99	<0.001
<i>Summer</i>	-0.029	0.050	-0.59	0.557	-0.035	0.050	-0.70	0.482
<i>Mining presence</i>	0.126	0.085	1.49	0.136	0.060	0.086	0.69	0.488
<i>Number of transactions</i>	0.008	0.002	3.51	<0.001	0.009	0.002	4.14	<0.001
<i>Flow $\times$ SD</i>	-0.025	0.005	-5.50	<0.001	-0.025	0.005	-5.42	<0.001
<i>Lagged price</i>	0.743	0.015	48.42	<0.001	0.718	0.016	44.77	<0.001
<i>Lagged SD</i>	0.486	0.088	5.52	<0.001	0.426	0.090	4.74	<0.001
<i>SPI-12</i>	0.102	0.044	2.34	0.020	0.290	0.102	2.85	0.004
<i>W $\times$ transaction flow</i>	—	—	—	—	-0.001	0.001	-0.71	0.478
<i>W $\times$ mining presence</i>	—	—	—	—	0.496	0.154	3.23	0.001
<i>W $\times$ number of transactions</i>	—	—	—	—	-0.010	0.004	-2.02	0.043
<i>W $\times$ flow $\times$ SD</i>	—	—	—	—	-0.014	0.008	-1.64	0.100
<i>W $\times$ lagged price</i>	—	—	—	—	0.112	0.042	2.68	0.007
<i>W $\times$ lagged SD</i>	—	—	—	—	0.064	0.156	0.41	0.681
<i>W $\times$ SPI-12</i>	—	—	—	—	-0.200	0.112	-1.78	0.075
<i>W $\times$ price (ρ)</i>	0.079	0.022	3.52	<0.001	-0.032	0.037	-0.86	0.389
<i>Log likelihood</i>	-2,620				-2,603			
<i>Akaike criterion</i>	5,261				5,241			
<i>Pseudo R²</i>	0.697				0.702			
<i>Observations</i>	1,785				1,785			

Supplementary Material

TABLE A1. Decomposition of the estimated effects into own-basin, spillover and total components. Direct, indirect and total effects computed from the spatial Durbin estimates in Table 2 following LeSage and Pace (2009). For covariate k the multiplier matrix is $S \sim k \sim (\mathbf{W}) = (\mathbf{I} - \rho \mathbf{W})^{-1} (\beta \sim k \sim \mathbf{I} + \theta \sim k \sim \mathbf{W})$; the direct effect is its mean diagonal, the total effect its mean row sum, and the indirect effect the difference. Effects are expressed in log price units. The final column reports the ratio of indirect to direct effects, which under the autoregressive restriction would equal $\rho/(1 - \rho)$ for every covariate and take the value 0.086. That the ratio instead varies from -1.05 to 8.46 and changes sign is a property of the Durbin specification.

Variable	Direct	90% Interval	Indirect	90% Interval
Lagged price	0.717	0.692 to 0.744	0.087	0.037 to 0.136
Lagged SD	0.428	0.281 to 0.575	0.050	-0.204 to 0.301
SPI-12	0.290	0.118 to 0.458	-0.203	-0.392 to -0.017
Mining presence	0.055	-0.090 to 0.199	0.480	0.229 to 0.729
Number of transactions	0.0095	0.006 to 0.013	-0.0098	-0.018 to -0.002
Summer	-0.035	-0.116 to 0.049	0.001	-0.002 to 0.006
Flow $\times$ SD	-0.025	-0.032 to -0.017	-0.013	-0.026 to 0.001
Transaction flow	-0.0009	-0.0013 to -0.0005	-0.0004	-0.0015 to 0.0006